space structures laboratory

Scaling Limits for Ring-Like Structures Robotically Assembled in Space

Tara Venkatadri and Sergio Pellegrino

SSDM 2026 Presentation

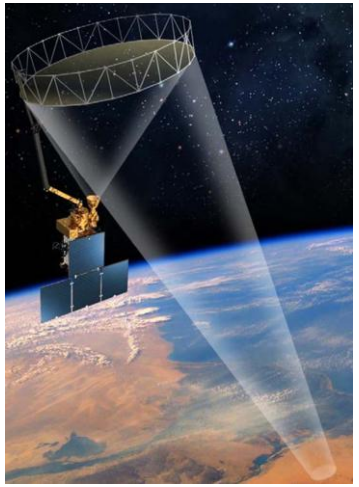
June 9th, 2026

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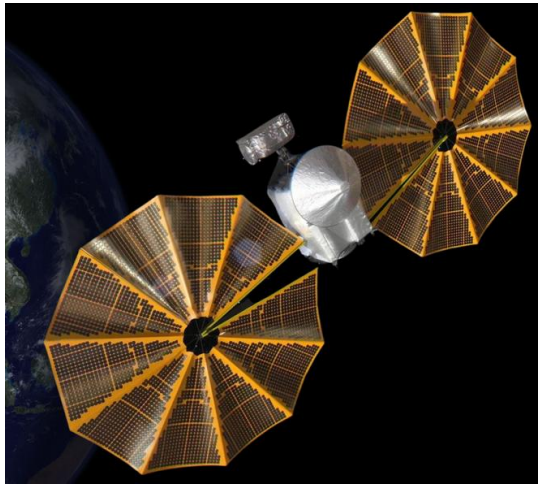
In-Space Assembly of Large Structures

- Growing need for large spacecraft structures
  - Improved performance (e.g. higher resolution and power generation)
- Large deployable structures exceed launch vehicle limits: Transport to space becomes challenging
- Solution: In-space assembly
  - Variety of proposed robotic methods in literature

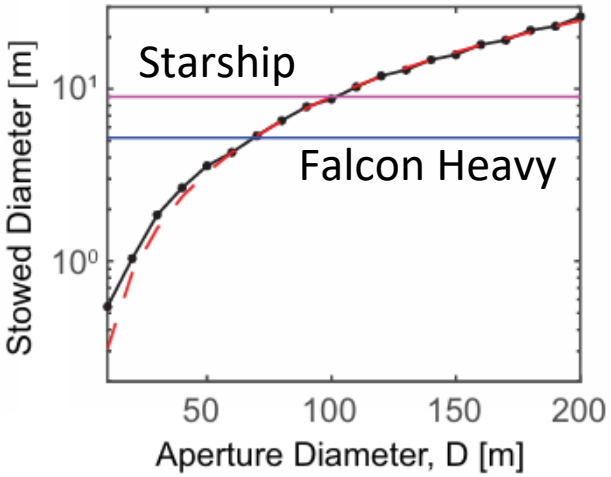
SMAP Antenna [1]



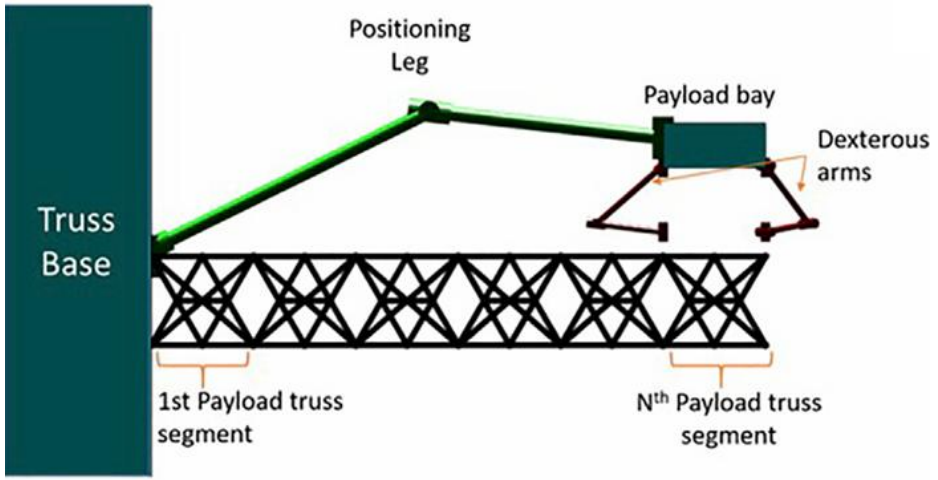
Lucy Solar Array [2]



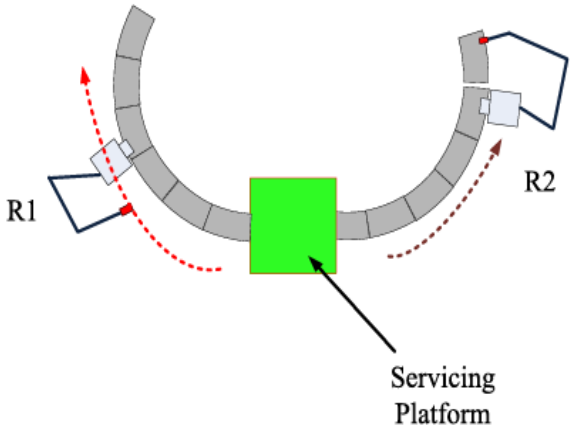
Deployable Reflector Scaling [3]



Long-Reach Robot Arm [4]



Robot Traversing Structure [5]



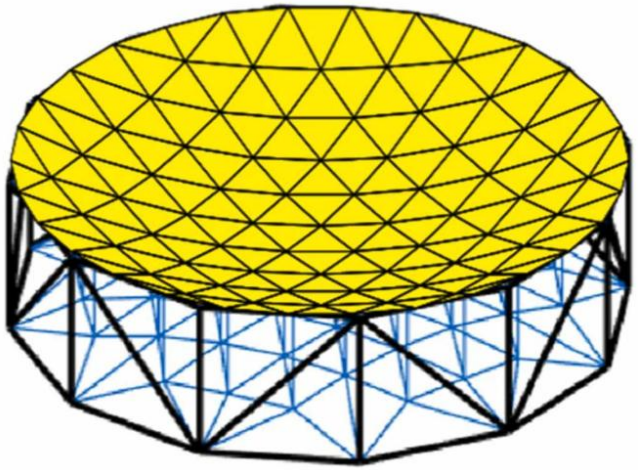
[1] Entekhabi, D. et al. (2010). The Soil Moisture Active Passive (SMAP) mission
[2] Lounsbury, J., & Sekerak, M. (2025). Post-launch verification of Lucy solar array deployment
[3] Suh, J. E., et al. (2024). Scaling laws for deployable mesh reflector antennas
[4] McBryan, K. (2022). Comparing multi-arm robotics for in-space assembly
[5] Cao, K., et al. (2021). Dynamics and on-orbit assembly strategies for an orb-shaped solar array

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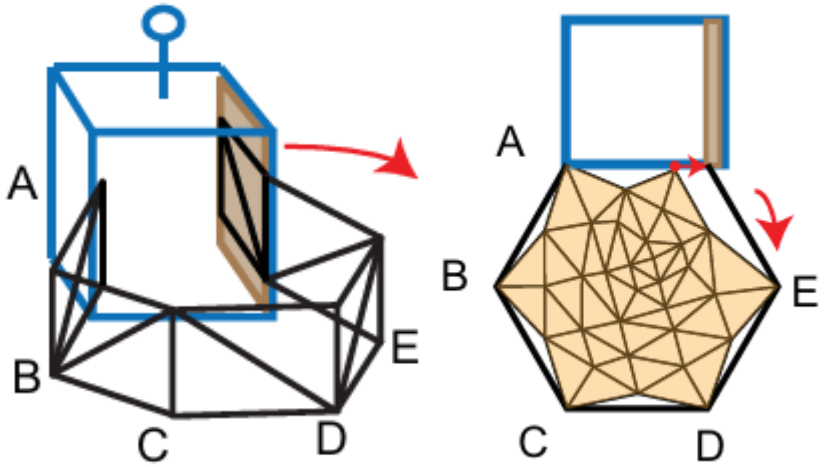
DARPA NOM4D Stationary Robot Scheme

- Many structures involve ring-like perimeter truss and inner cable-supported membrane
- Stationary robot method: Bay assembly, cable attachment, ejection from truss builder on movable assembly plate
- Recent application to 12-sided AstroMesh-style reflector prototype

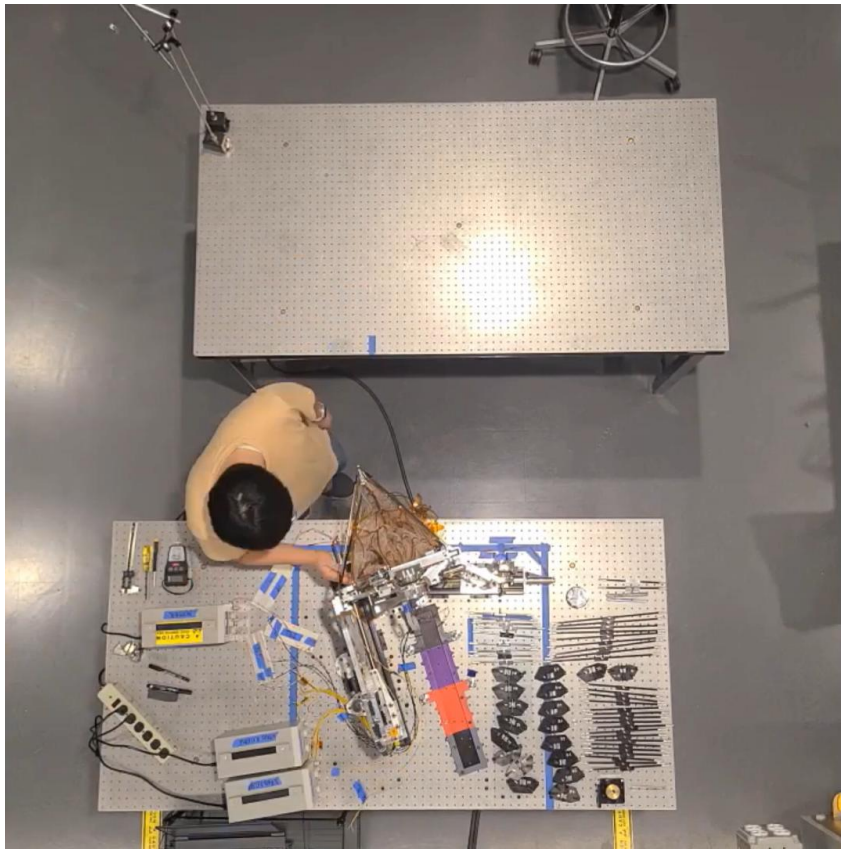
AstroMesh Structure [6]



Stationary Robot [7]



12-Sided Reflector Prototype

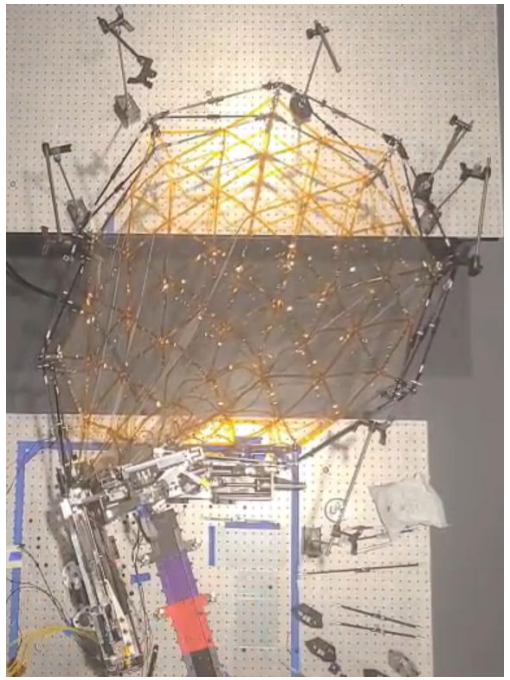


[6] Zhang, Y., et al. (2024). Design of deployable mesh reflector antenna based on cable-dome tensegrity structure
[7] Suh, J. E., et al. (2024). Concept for scalable mesh reflector antennas assembled in space

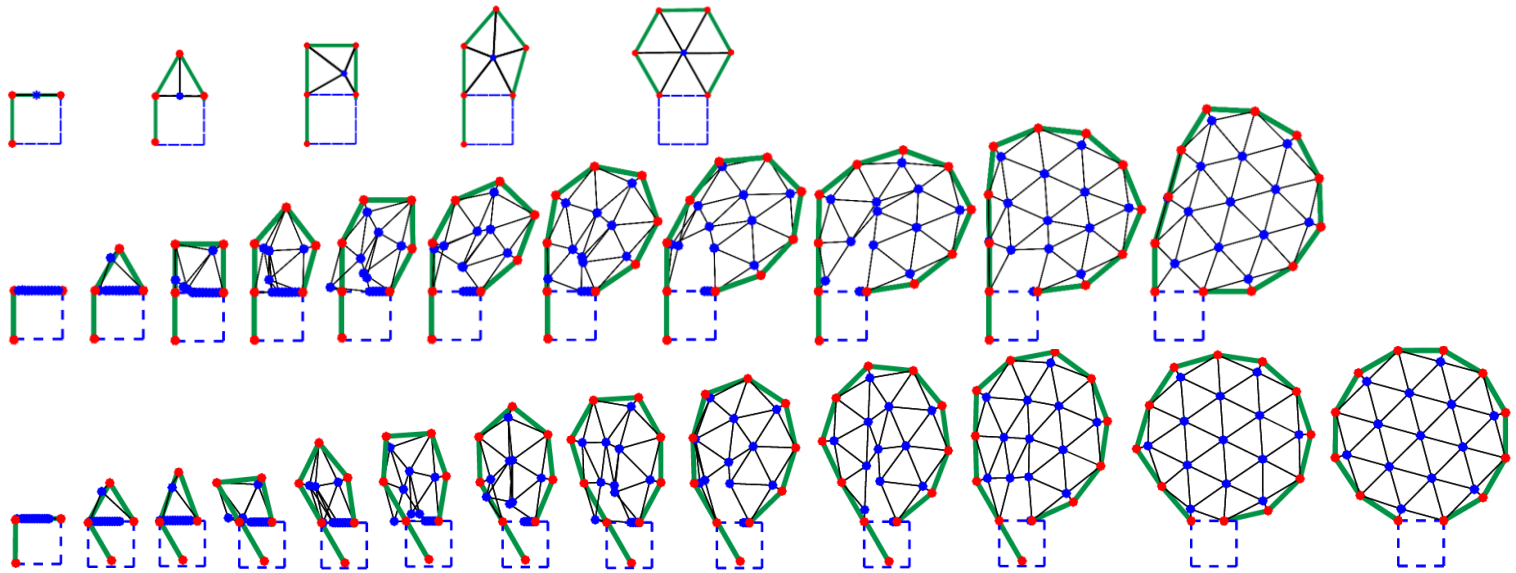
Numerical Model of Assembly Process

- Stepwise shapes unknown due to lack of rigidity in intermediate configurations
- Kinematic constraints affect feasibility (may cause jamming in intermediate shapes)
  - Need a mathematical model to predict assembly feasibility
- Prior work: Finite element model (FEM) of AstroMesh reflector assembly
  - Sensitivity to constraints increases with size of structure

Slack Cables at Early Steps



AstroMesh Assembly FEM [8]

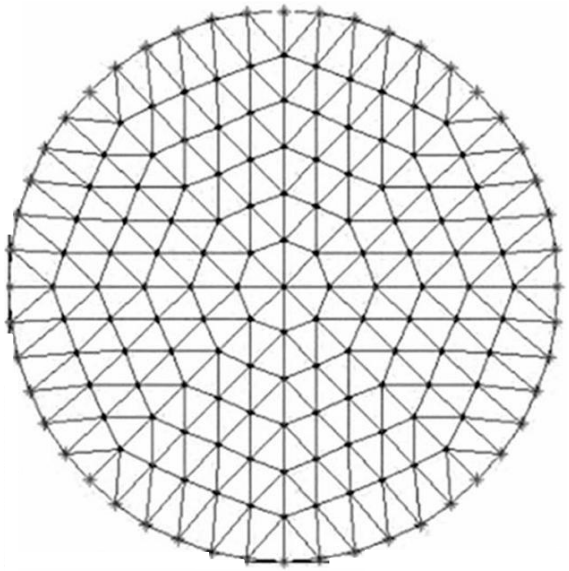


[8] Dassanayake, S. P., et al. (2025). Numerical study of novel concept for in-space assembly of ring-like space structures

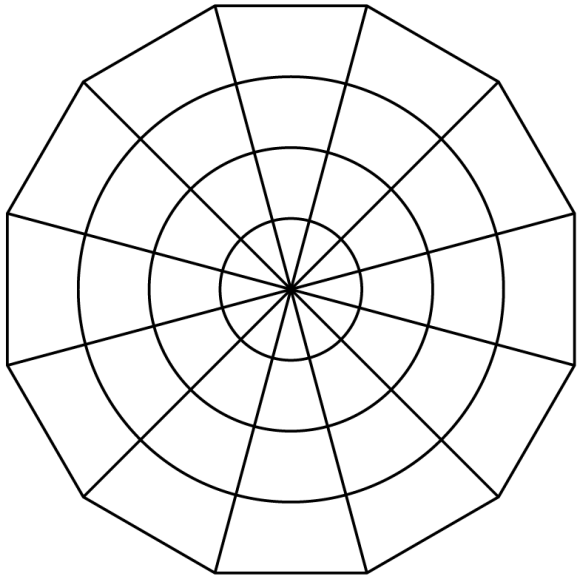
New Applications of Stationary Robot

- Current focus: Expansion beyond 12-sided AstroMesh reflector
  - Larger structures for which in-space assembly is necessary
  - Prismatic ring-like structures with different cable topologies
- Existing FEM is highly specialized: Cannot be applied to other structures
  - Need a new model to test scaling and alternative use cases of stationary robot
- Use diagonal cable topology to initialize model and study assembly of larger structures

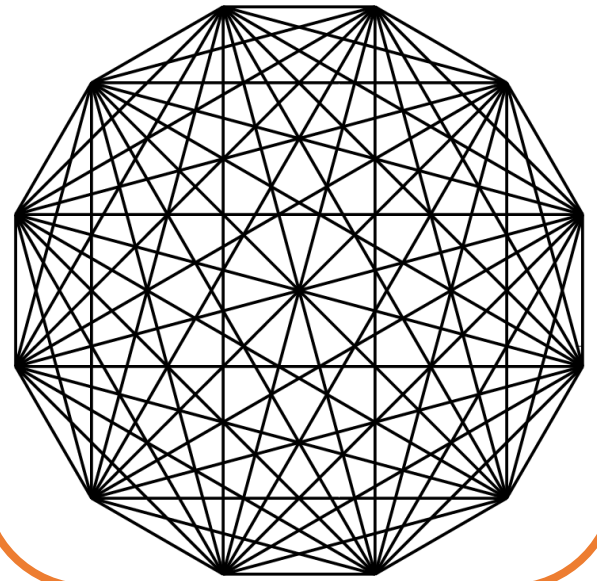
Pseudogeodesic Cables [9]



Radial and Circumferential Cables [6]



Diagonal Cables



[6] Zhang, Y., et al. (2024). Design of deployable mesh reflector antenna based on cable-dome tensegrity structure
[9] Shi, H., et al. (2018). New methodology of surface mesh geometry design for deployable mesh reflectors

Outline

Goal

Extend application of stationary robot assembly method to increasingly large regular polygons with diagonal cables

Formulate mathematical problem of diagonal cable assembly with stationary robot

Demonstrate analytical solution for assembling structures of any size

Construct numerical optimization model to account for practical limitations

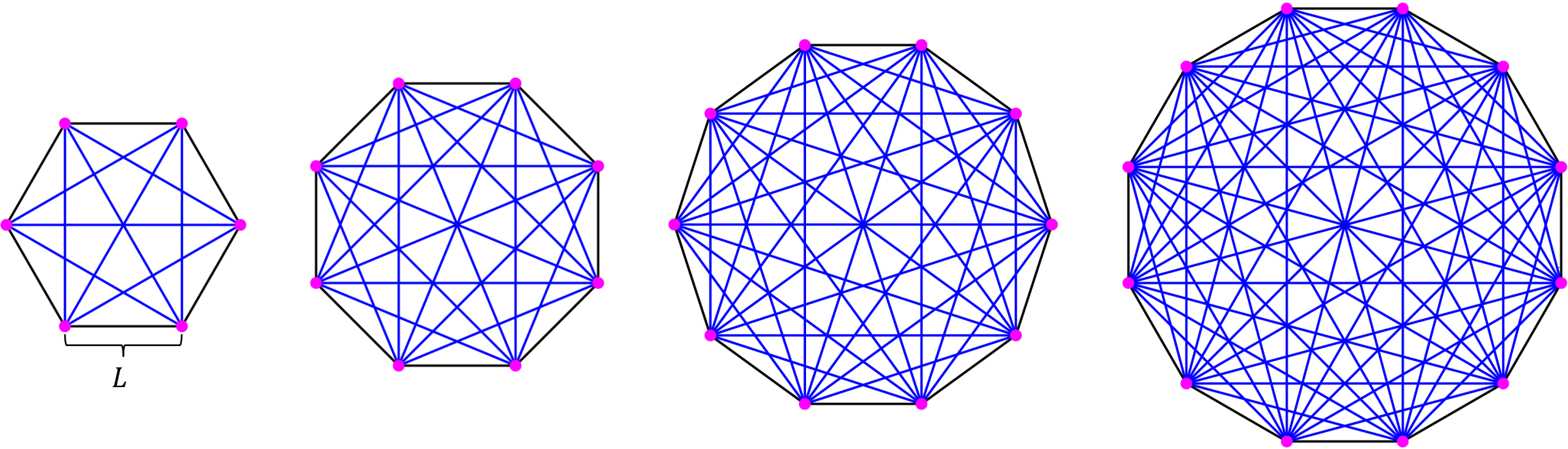
Illustrate effect of kinematic constraints on assembly feasibility

Identify scaling limits as structure's size increases

Diagonal Cable Topology

- Struts form n-sided regular polygon on perimeter of 2D structure
- For fixed side length L, larger structures can be produced by increasing n
- Cables connect each pair of non-adjacent vertices: Mathematically simple to define
  - More practical to use a subset of cables to reduce overlap
- Solutions for full set of cables guarantee feasible assembly with any subset of cables

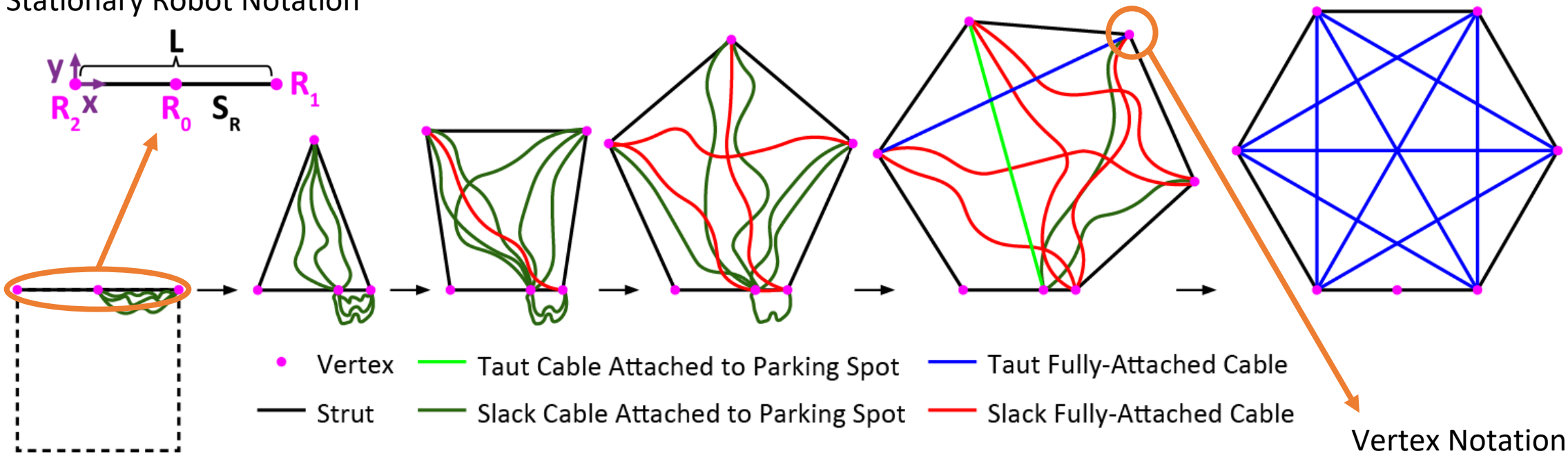
$n = 6$ $n = 8$ $n = 10$ $n = 12$



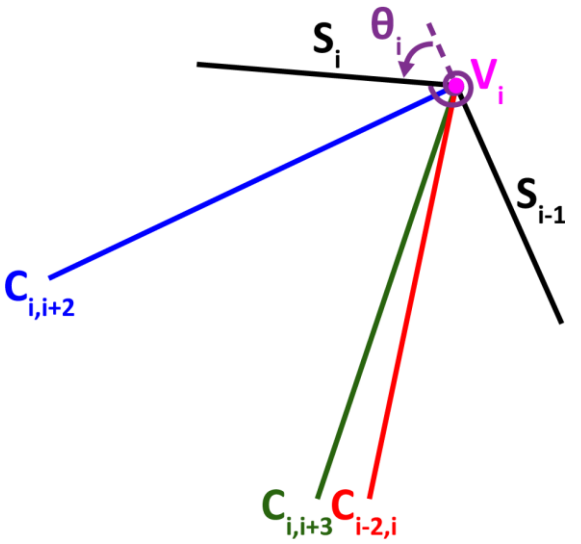
Conceptual Features of Model

Stationary Robot Notation

Hexagon Assembly Sequence



Vertex Notation



- Key assumption: Discrete strut ejection
- Main parameters: Distances of R_0 and R_1 from R_2 on x-axis
- Vertices V_i, rigid struts $\vec{S}_i$, inextensible cables $\vec{C}_{r,s}$
- Torsional spring angles θ_i, convexity constraint enforcing $\theta_i \geq 0$

Regular Polygon Feasibility

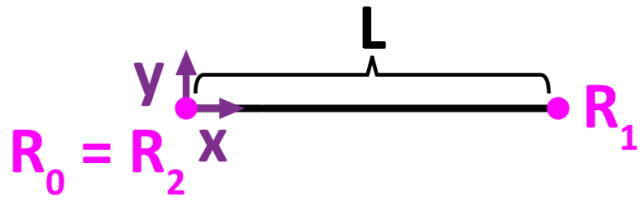
- Simplest solution: $R_0 = 0$ (coinciding with R_2), $R_1 = L$
- For octagon, symmetric geometry produces regular polygons of increasing size with all cables slack until final step
  - Intuitive method of producing desired final structure

Claim

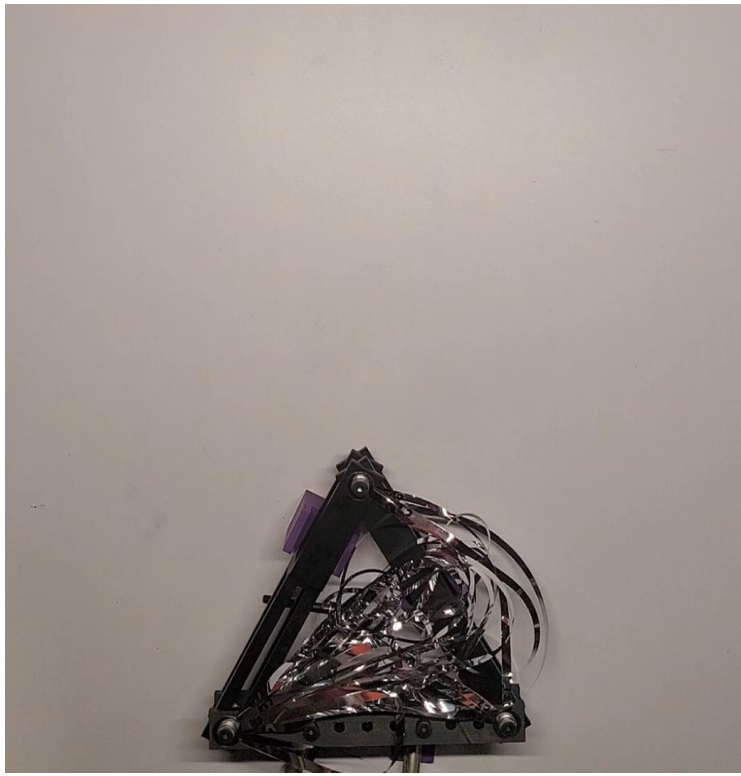
$(R_0, R_1) = (0, L)$ provides a feasible assembly process for any n

- Constraints automatically satisfied: Strut lengths, convexity, cable lengths in final configuration
- Constraints to examine: Cable lengths in intermediate configurations (fully-attached and attached to R_0)

Positions of R_0 and R_1



Octagon Prototype



Regular Polygon Feasibility

Fully Attached Cables

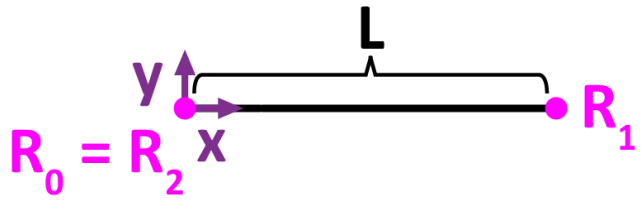
- Length of cable with k-vertex separation in q-sided intermediate polygon:
  - $c(q,k,L) = \frac{L\sin(\pi k/q)}{\sin(\pi/q)}$
- As assemb… and c incr…
  - **Interm…**

Cables Attached to R_0

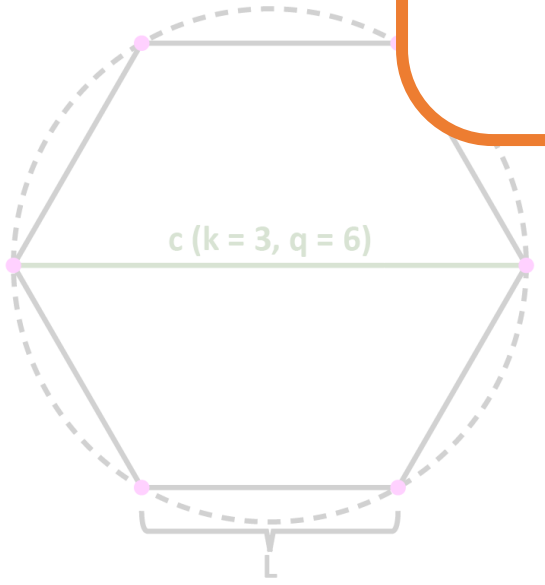
- $\vec{C}_{r,s}$ vertex separation $\leq k$ at earlier steps
- Cable with $(k-1)$-vertex separation:
  - $\cos(\pi/q)$
  - $q)$
- …creased vertex …e length
- **taut length**
- …with $k = 3$ in …l Polygon

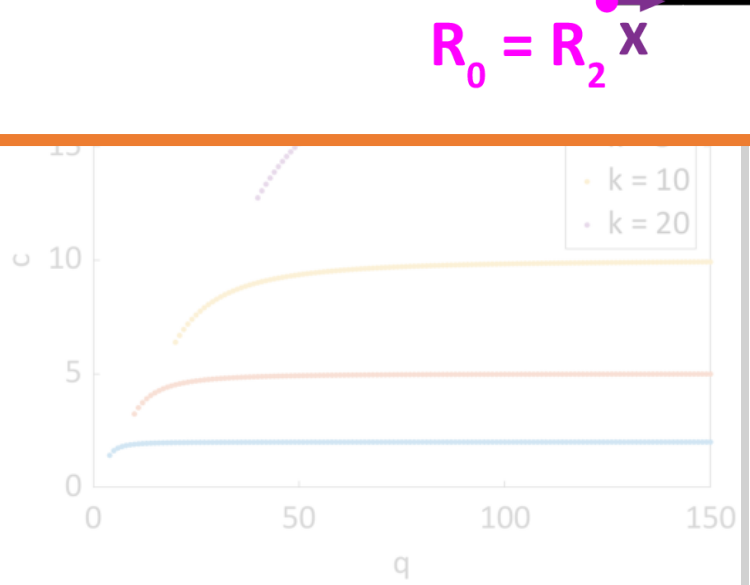
Conclusion

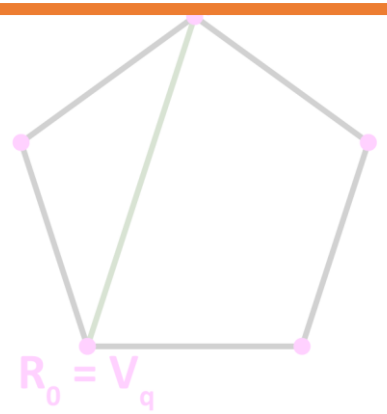
$(R_0, R_1) = (0, L)$ provides a feasible assembly process consisting of regular polygons for any n

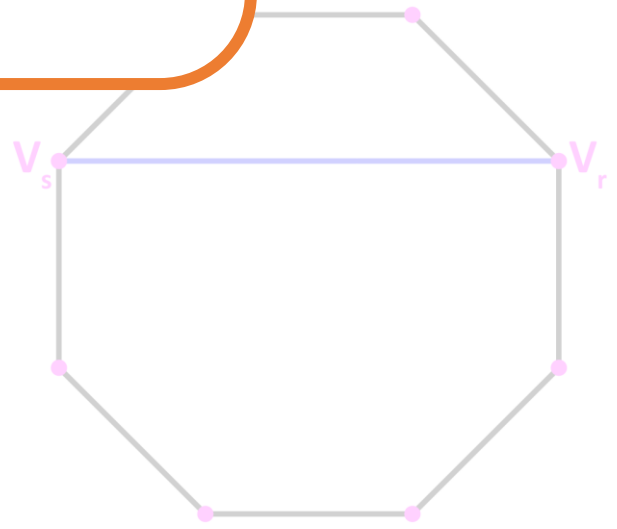


Fully-Attached Ca…



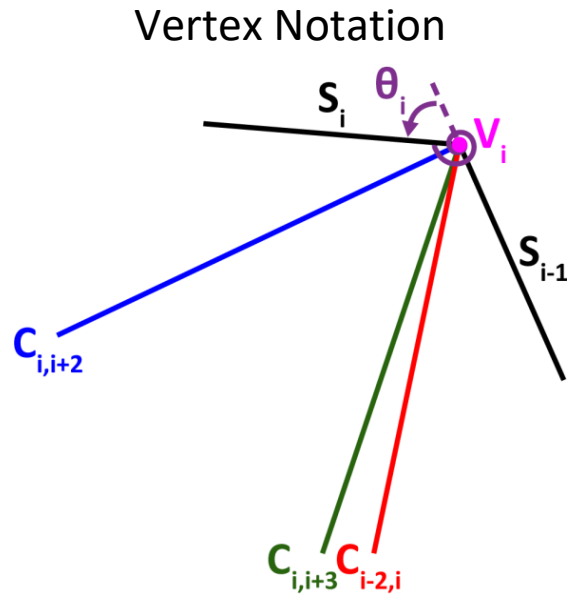






Numerical Optimization Model

- Practical issue with $R_0 = 0$: Overcrowding of strut ejection area
- Need $R_0 > 0$: Introduces asymmetry, requires numerical model
- Fundamental unknowns: Positions of vertices V_i
  - Struts: $\vec{S}_i = \begin{pmatrix} V_{i+1,x} - V_{i,x} \\ V_{i+1,y} - V_{i,y} \end{pmatrix}$
  - Cables: $\vec{C}_{r,s} = \begin{pmatrix} V_{s,x} - V_{r,x} \\ V_{s,y} - V_{r,y} \end{pmatrix}$
  - Torsional spring angles: $\theta_i = \arccos\left(\frac{\vec{S}_i \cdot \vec{S}_{i-1}}{|\vec{S}_i||\vec{S}_{i-1}|}\right)$



Optimization Framework

- Objective: Minimize torsional spring strain energy, $\frac{1}{2}\alpha \sum_{i=1}^{n} \theta_i^2$
- Constraints: $\left|\vec{S}_i\right| = L$, $\left|\vec{C}_{r,s}\right| \leq L_C$, $\theta_i \geq 0$

Numerical Implementation

- Algorithm: MATLAB fmincon SQP [10]
- Guesses: Preserve sequential assembly, promote smooth convergence, add 1 vertex at each step

[10] Nocedal, J., & Wright, S. J. (2006). Numerical optimization

Octagon Assembly Sequence

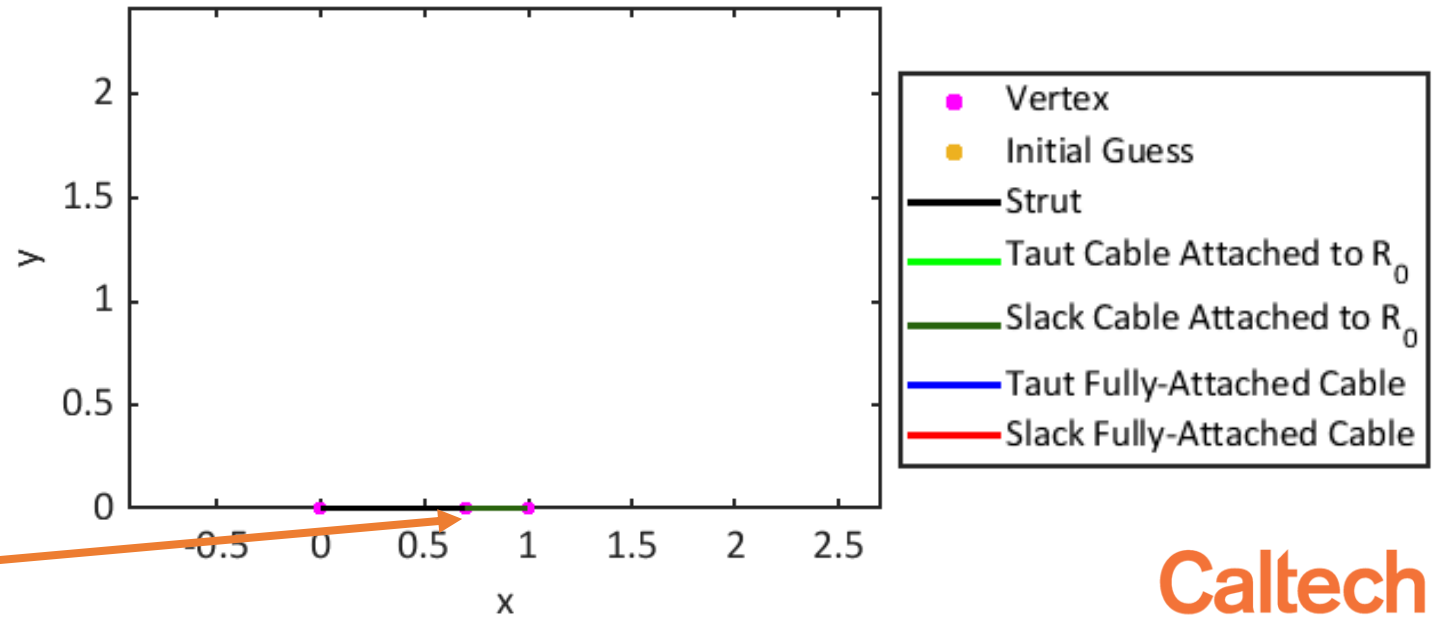
- Focus on $R_1 = L$ to isolate effect of R_0
- If requiring $R_0 > 0$, need $R_0 \geq 0.623L$ to prevent infeasible optimization due to extension of cables beyond taut lengths

Prototype: $R_0 = 0.7L$



R_0

MATLAB Model: $R_0 = 0.7L$



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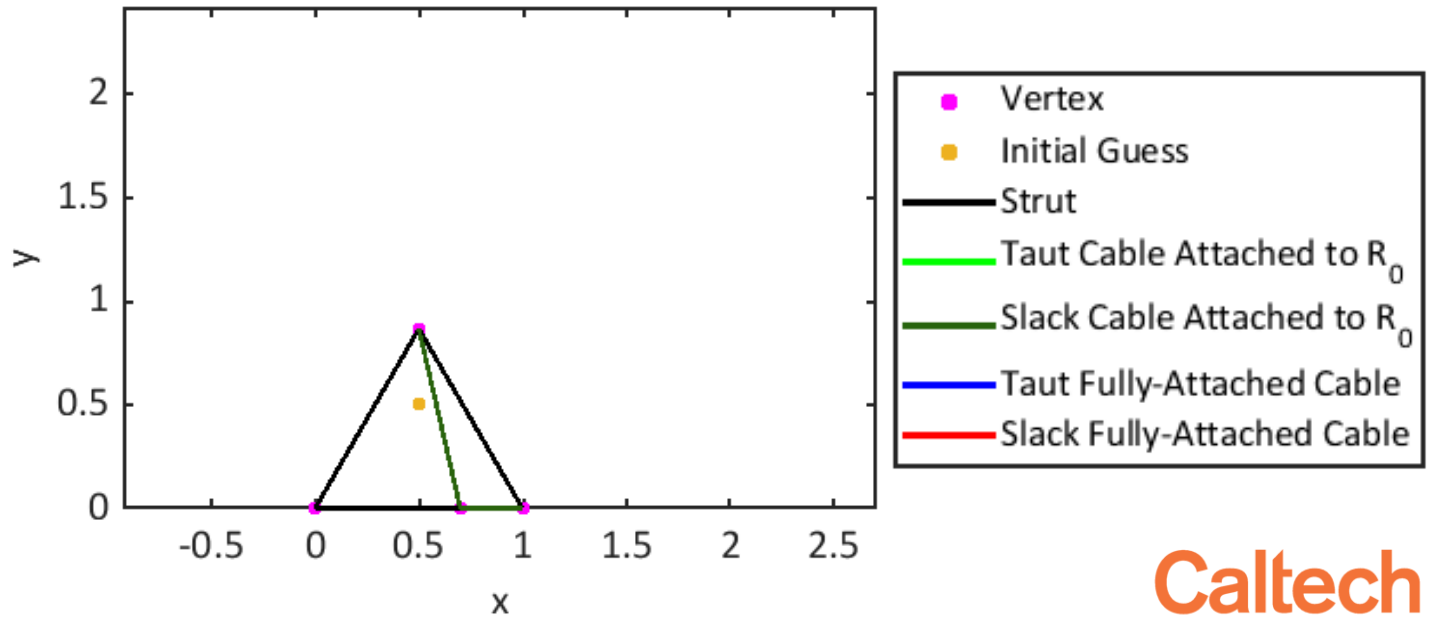
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Octagon Assembly Sequence

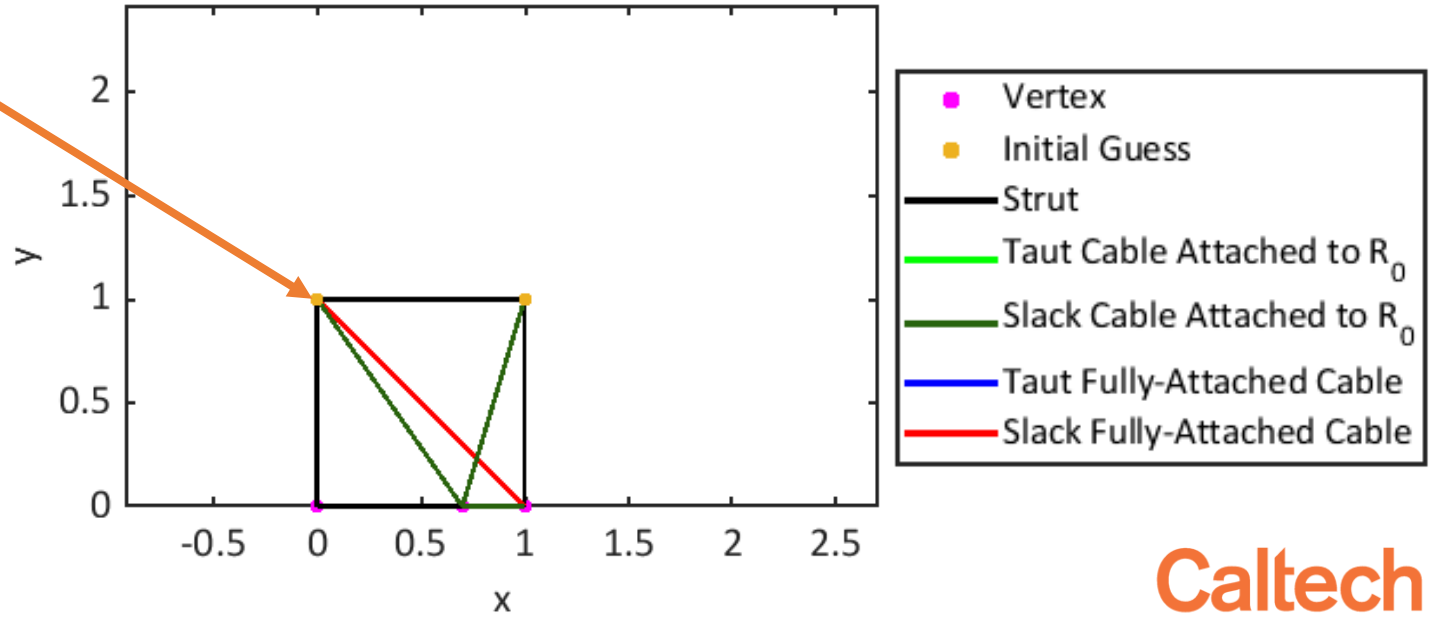
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Regular polygon due to slack cables

MATLAB Model: $R_0 = 0.7L$

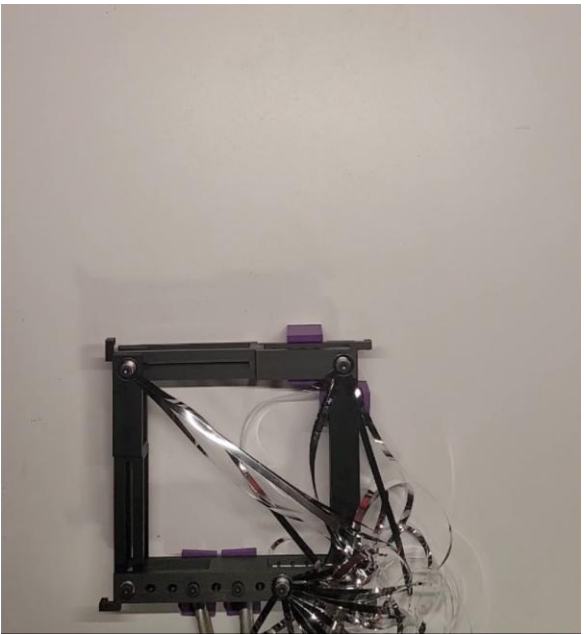


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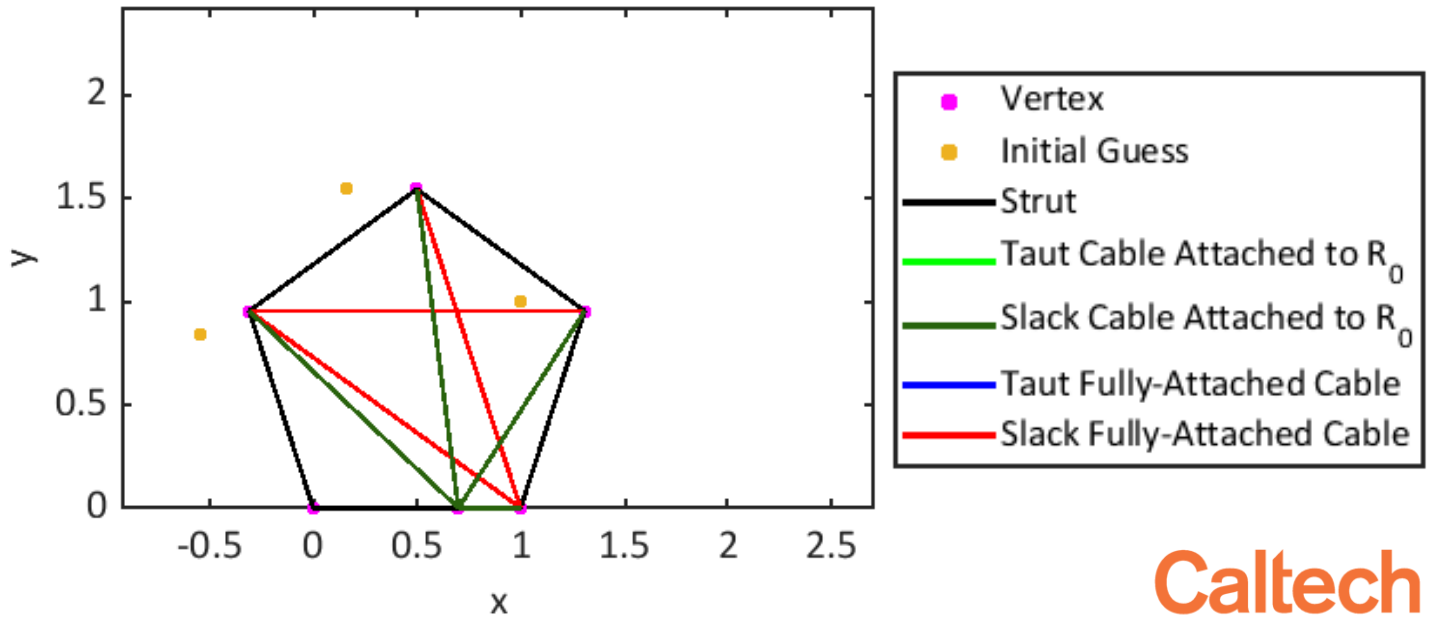
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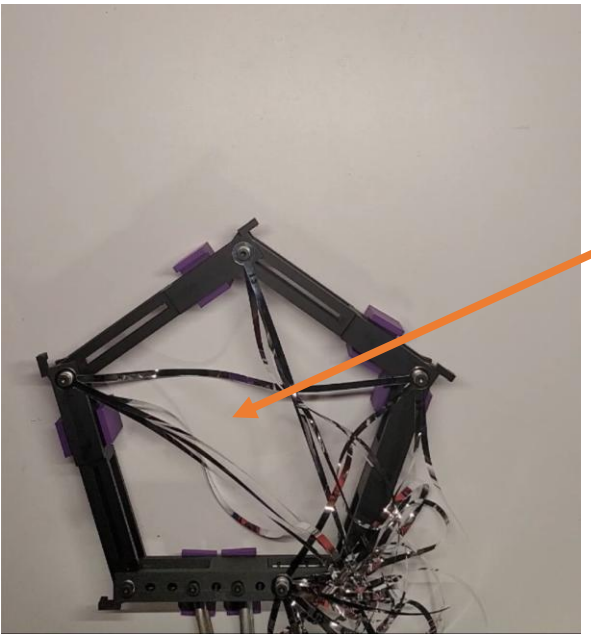


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Octagon Assembly Sequence

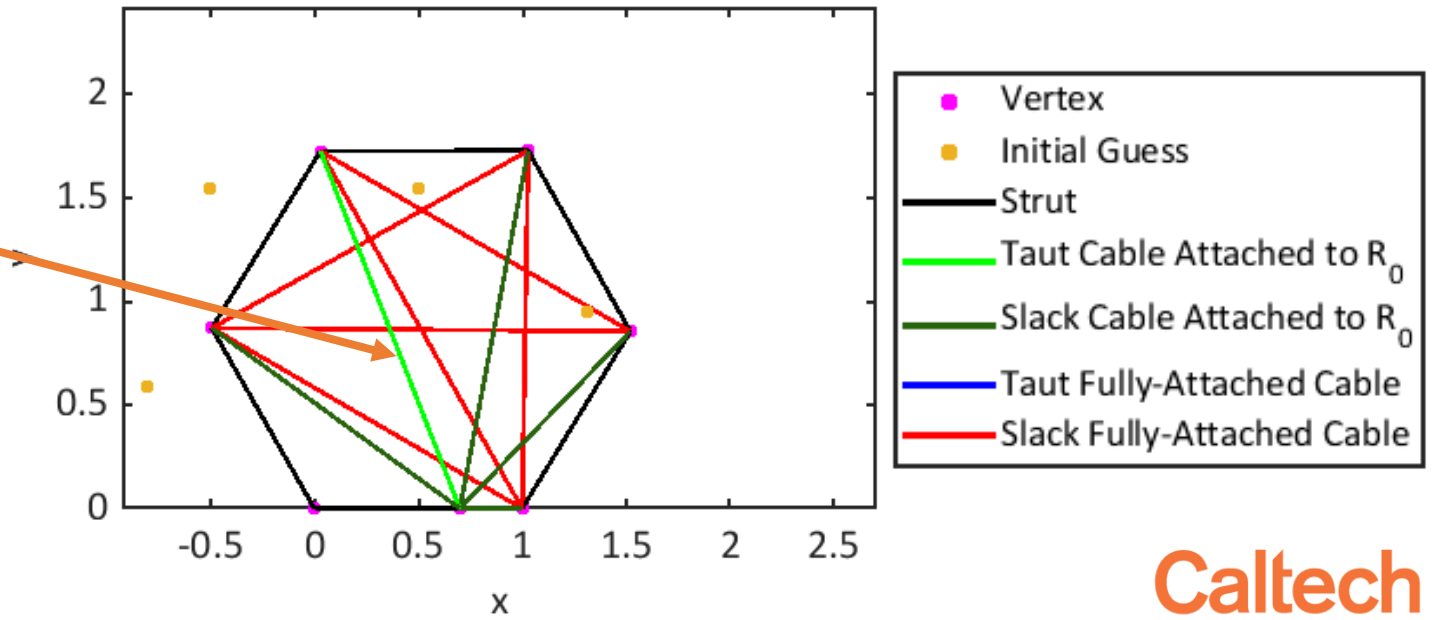
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- If requiring $R_0 > 0$, need $R_0 \geq 0.623L$ to prevent infeasible optimization due to extension of cables beyond taut lengths

Prototype: $R_0 = 0.7L$



Irregular shape due to taut cables

MATLAB Model: $R_0 = 0.7L$



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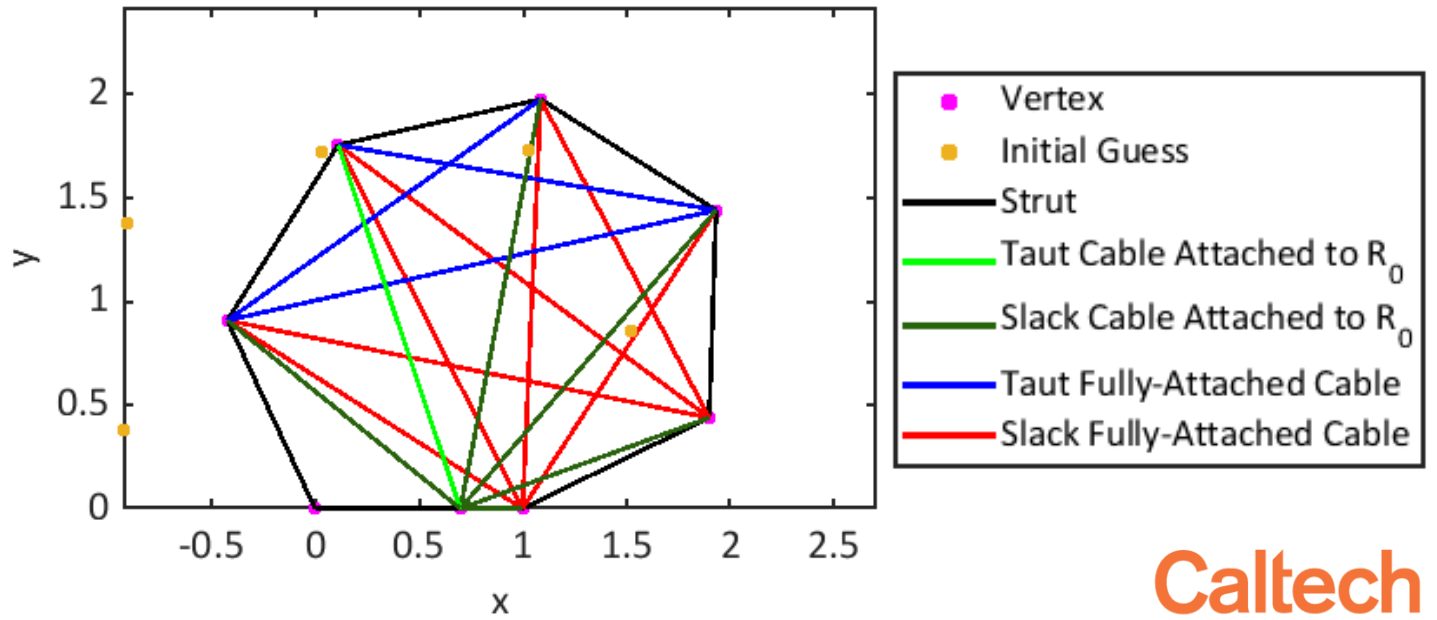
Octagon Assembly Sequence

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Prototype: $R_0 = 0.7L$



MATLAB Model: $R_0 = 0.7L$



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Octagon Assembly Sequence

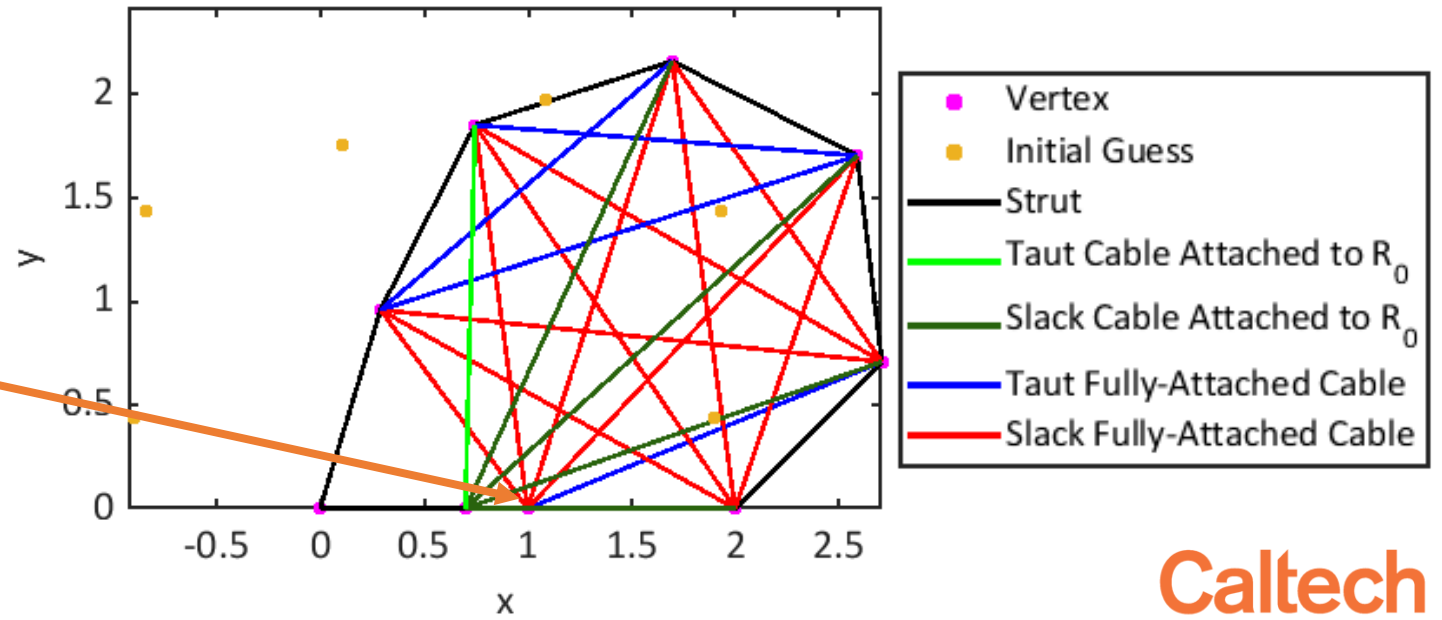
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Prototype: $R_0 = 0.7L$



Angle stops active

MATLAB Model: $R_0 = 0.7L$

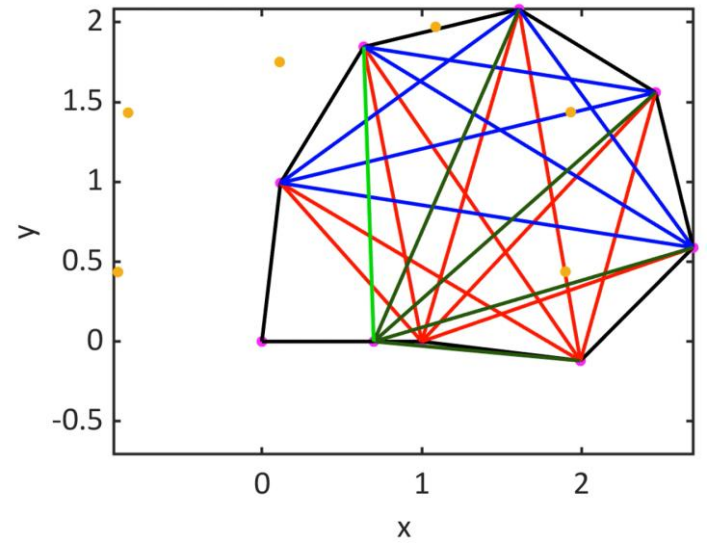


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Octagon Assembly Sequence

- Focus on $R_1 = L$ to isolate effect of R_0
- If requiring $R_0 > 0$, need $R_0 \geq 0.623L$ to prevent infeasible optimization due to extension of cables beyond taut lengths
- Without angle stops, penultimate and final shapes are concave (desired final structure is not obtained)
- Shows effect of kinematic constraints on assembly feasibility

Results Without Convexity Constraint

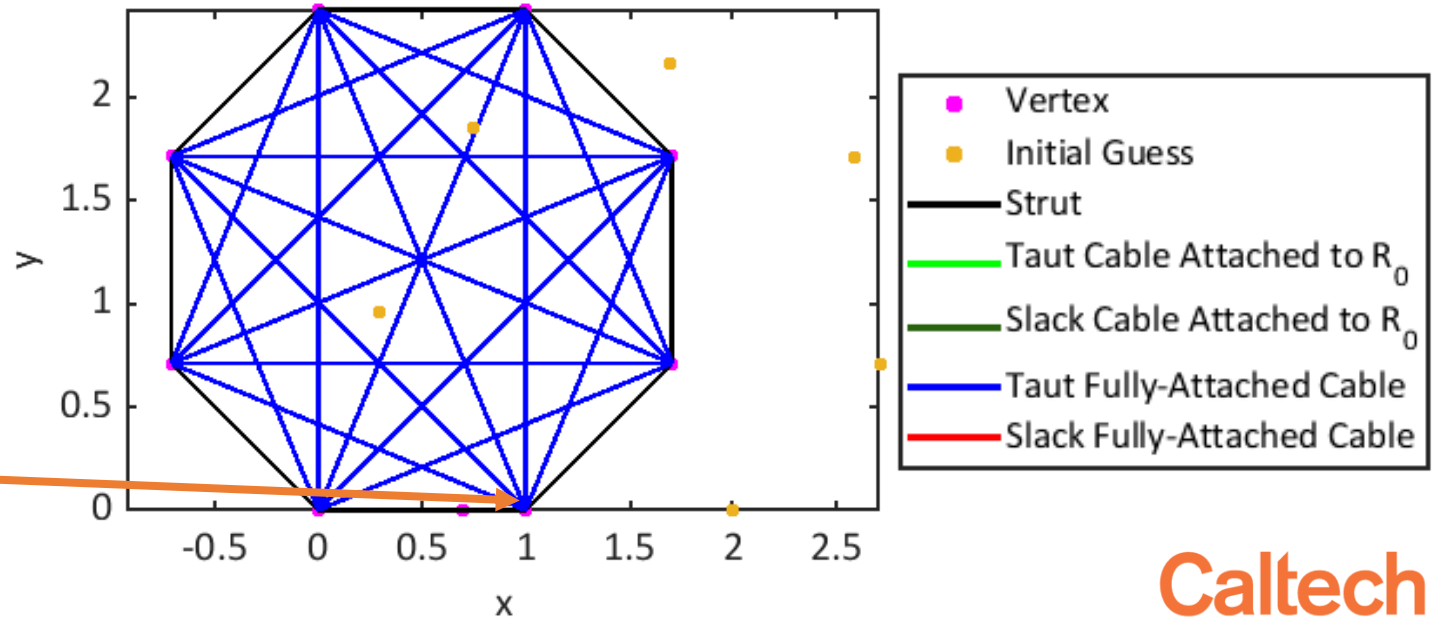


Prototype: $R_0 = 0.7L$



Desired final shape

MATLAB Model: $R_0 = 0.7L$

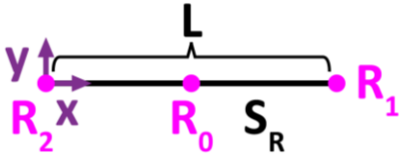


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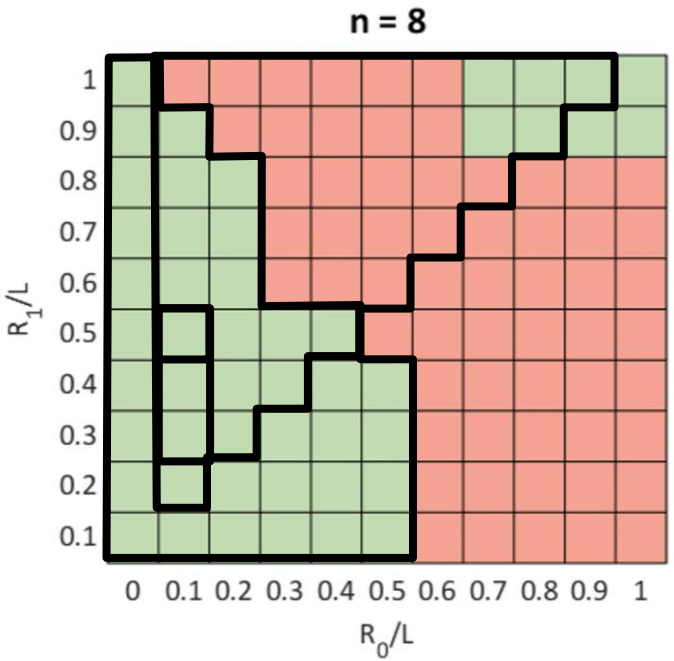
Scaling Study

- $R_1 = L$: Infeasible for any $R_0 > 0$ for $n \geq 9$, need to move R_1
- Performed (R_0, R_1) grid searches to establish assembly feasibility
  - Cases with $0 < R_0 < R_1$ are most practical for implementation
  - Range of valid solutions (and maximum R_0) shrinks with staircase structure as n increases
  - $n = 23$: Maximum for which $R_0 = 0.1L$ provides valid solutions

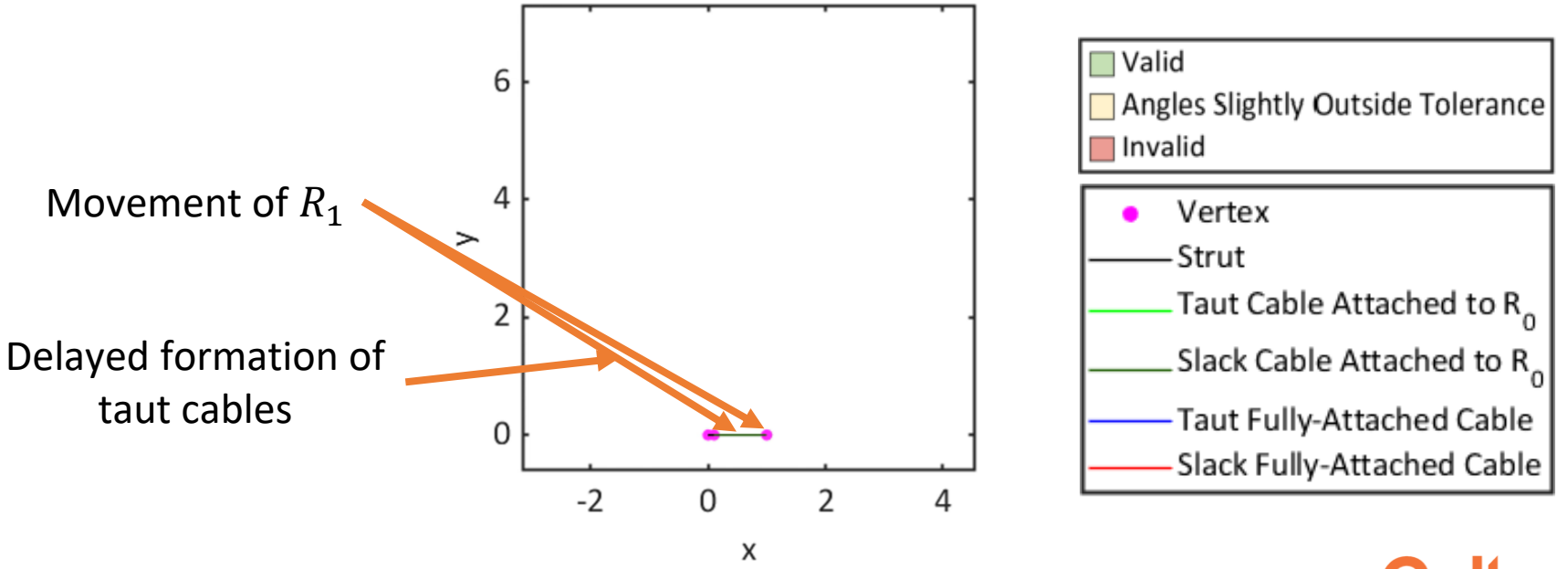
Stationary Robot Notation



Grid Searches for Varying n



Assembly Sequence: $n = 23$, $R_0 = 0.1L$, $R_1 = 0.5L$



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Conclusion

Summary

Applied stationary robot method to a new cable topology and demonstrated successful assembly of larger structures

Formulated mathematical assembly model for n-sided polygons with diagonal cables

Demonstrated analytical solution for assembling structures with any n if $R_0 = 0$

Constructed MATLAB numerical optimization model to study $R_0 > 0$

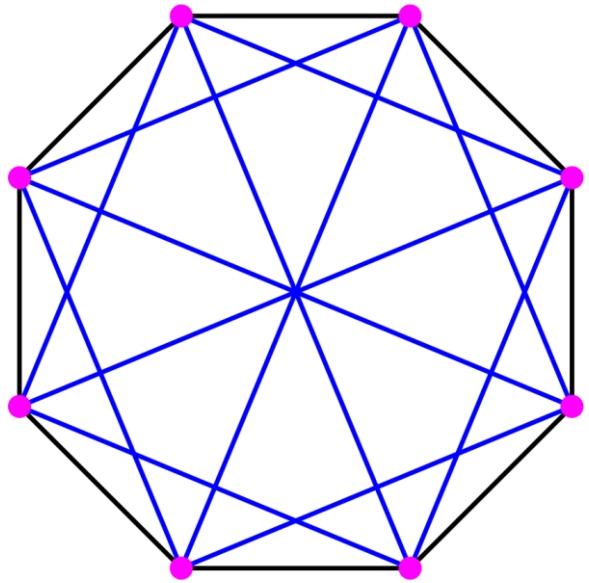
Illustrated effect of (R_0, R_1) and convexity constraint on assembly feasibility

Identified 23 sides as scaling limit with $R_0 > 0$ for given topology/model assumptions

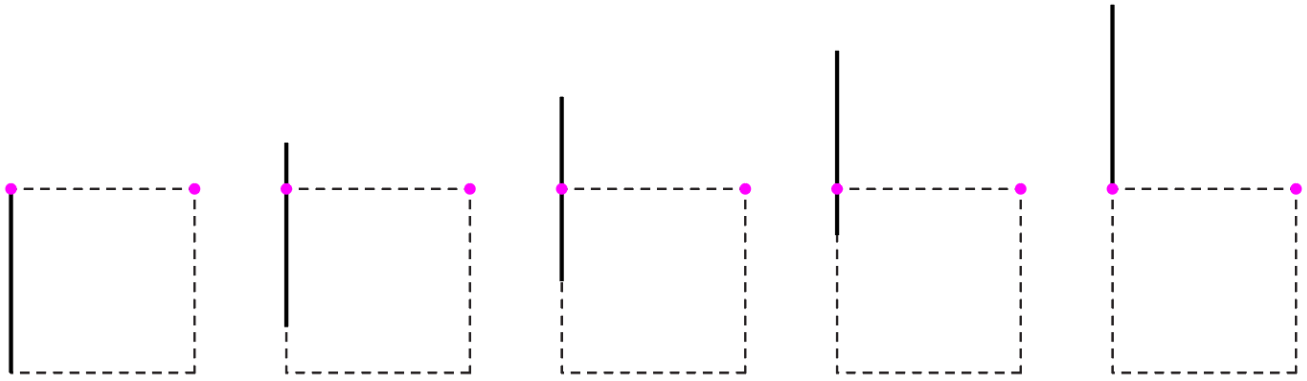
Future Work

- Adjust model to enable assembly of structures with $n > 23$:
  - Consider structures with a subset of cables removed
  - Determine if $R_0 \in (0, 0.1L)$ can be physically implemented
- Incorporate practical aspects of assembly not yet considered
  - Assembly plate ejection angle, continuous ejection
- Study attachment of planar functional membrane to cable support structure [11]

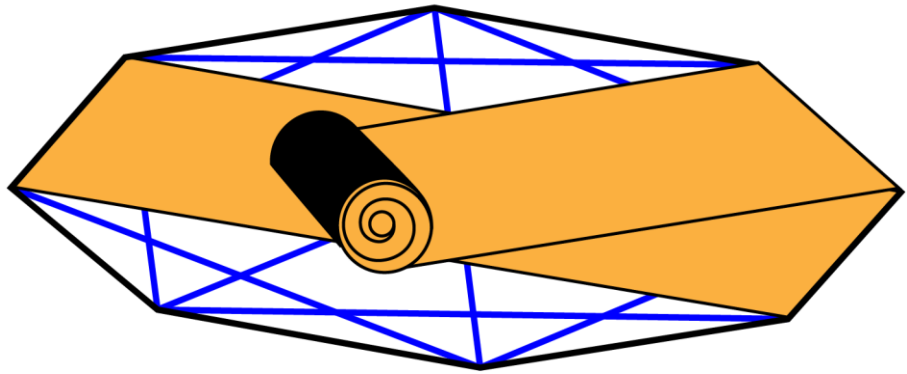
Octagon with Reduced Cables



Incremental Strut Ejection with Vertical Assembly Plate



Membrane Attachment



[11] Pierreux, B. G. H. (2023). Design and demonstration of autonomous placement of reflective panels for in-space antenna construction

Acknowledgements

- Financial support:
  - DARPA NOM4D Program
  - NSF Graduate Research Fellowship Program
- Development of stationary robot concept:
  - Dr. Jong-Eun Suh
  - Dr. Sahangi Dassanayake

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10. Nocedal, J., & Wright, S. J. (2006). *Numerical optimization* (2nd ed.). New York, N.Y. Springer.
11. Pierreux, B. G. H. (2023). *Design and demonstration of autonomous placement of reflective panels for in-space antenna construction* (Publication No. 23-029) [Master's thesis, ETH Zurich].

Conclusion

Summary

Applied stationary robot method to a new cable topology and demonstrated successful assembly of larger structures

Formulated mathematical assembly model for n-sided polygons with diagonal cables

Demonstrated analytical solution for assembling structures with any n if $R_0 = 0$

Constructed MATLAB numerical optimization model to study $R_0 > 0$

Illustrated effect of (R_0, R_1) and convexity constraint on assembly feasibility

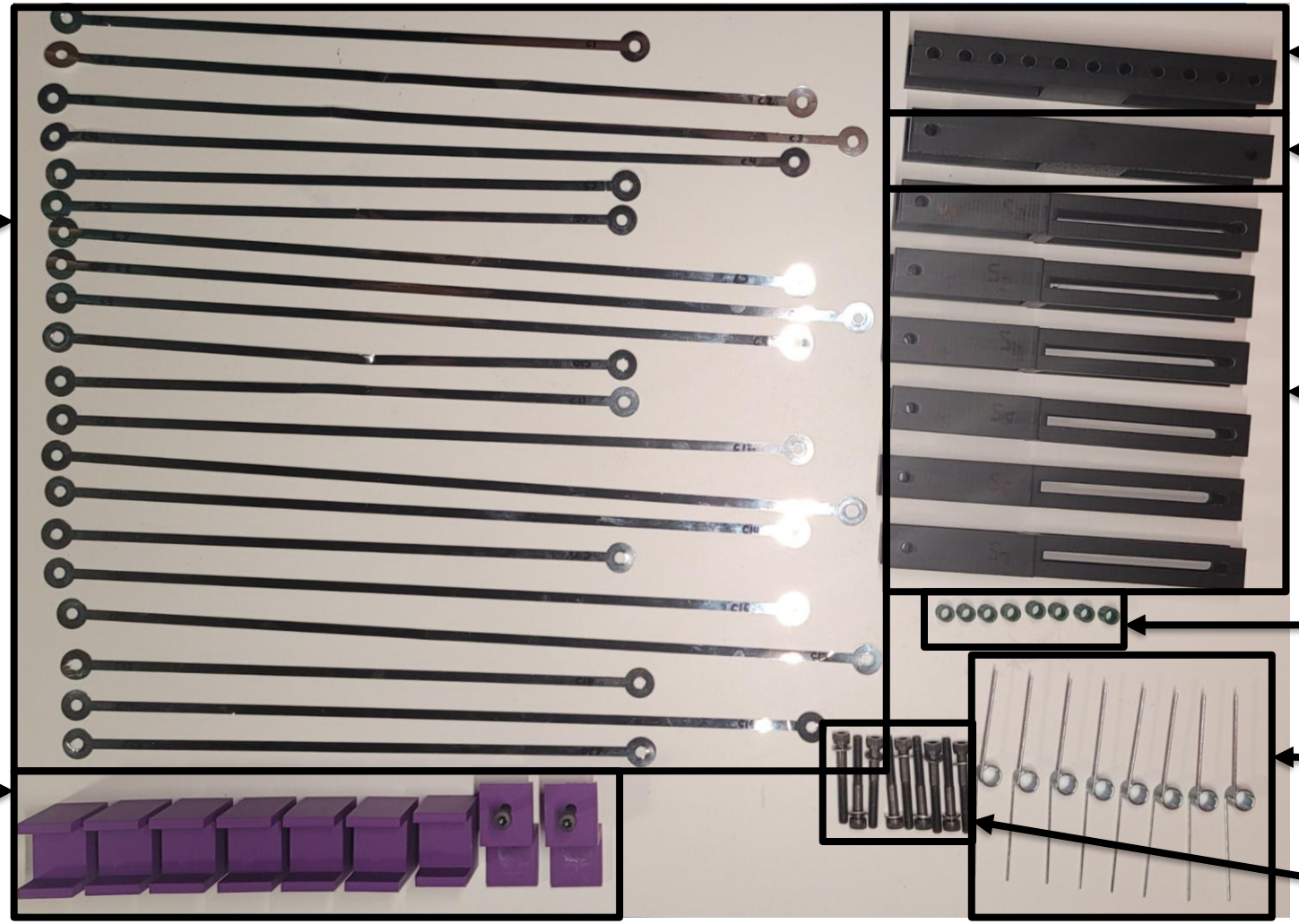
Identified 23 sides as scaling limit with $R_0 > 0$ for given topology/model assumptions

space structures laboratory

Backup Slides

Caltech

Prototype Components



Cables in 3 sizes to connect across 2-, 3-, and 4-vertex separations

Supports to raise model off table, for easier access to bolt holes during assembly

S_R with bolt holes for varying R_0

S_1 (constructed specially to interface with S_R)

Remaining struts

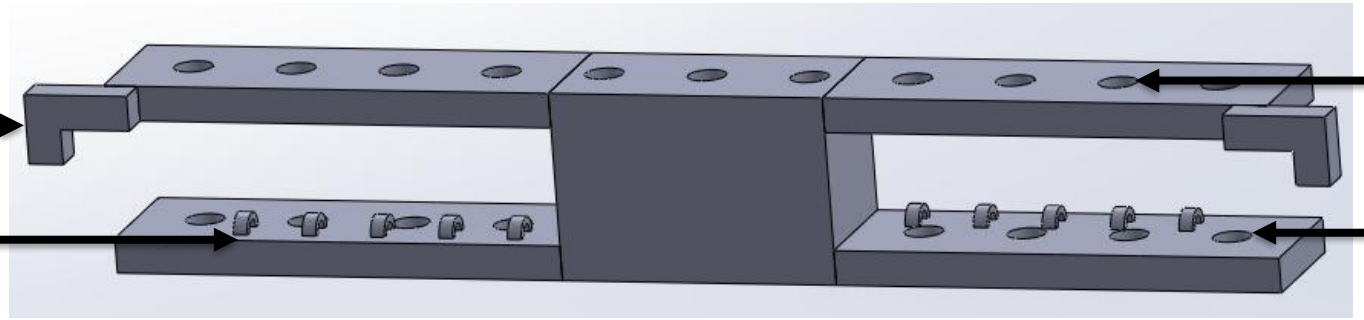
Collars inserted into torsional springs to ensure a tight fit with bolts

Torsional springs

Bolts with washers to secure torsional springs at joints

Prototype Strut Design

S_R



Angle stop (optional)

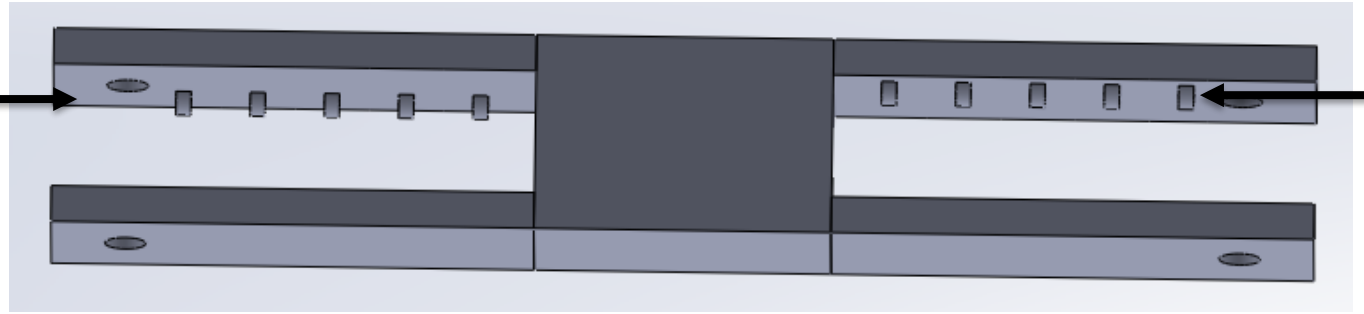
Loops to secure torsional spring end pieces in place

Bolt holes every $0.1L$ for varying R_0 placement

Bolt hole to secure torsional spring

S_1

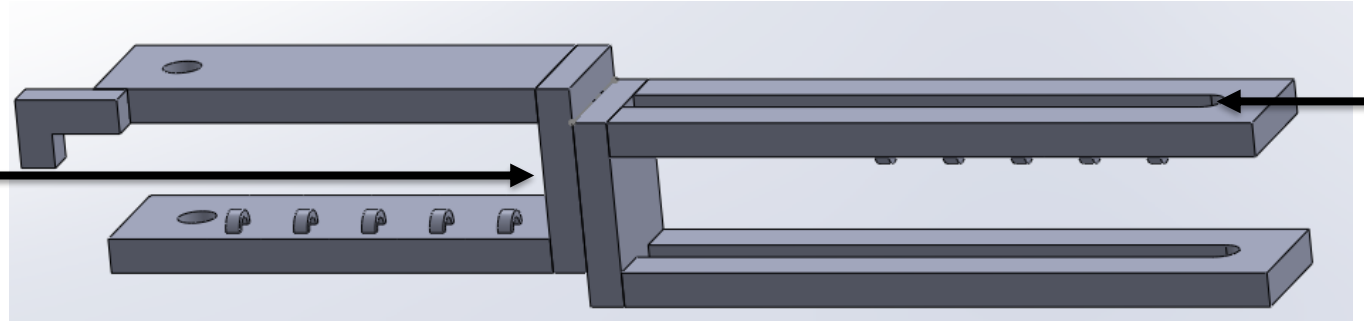
No need for angle stops due to alternating high-low strut positioning (S_1 is fully in low position, angle stops are on high-position struts)



Torsional spring loops on top arm of strut due to low positioning

Remaining Struts

Staggered high-low design to enable assembly of any number of struts (odd or even)

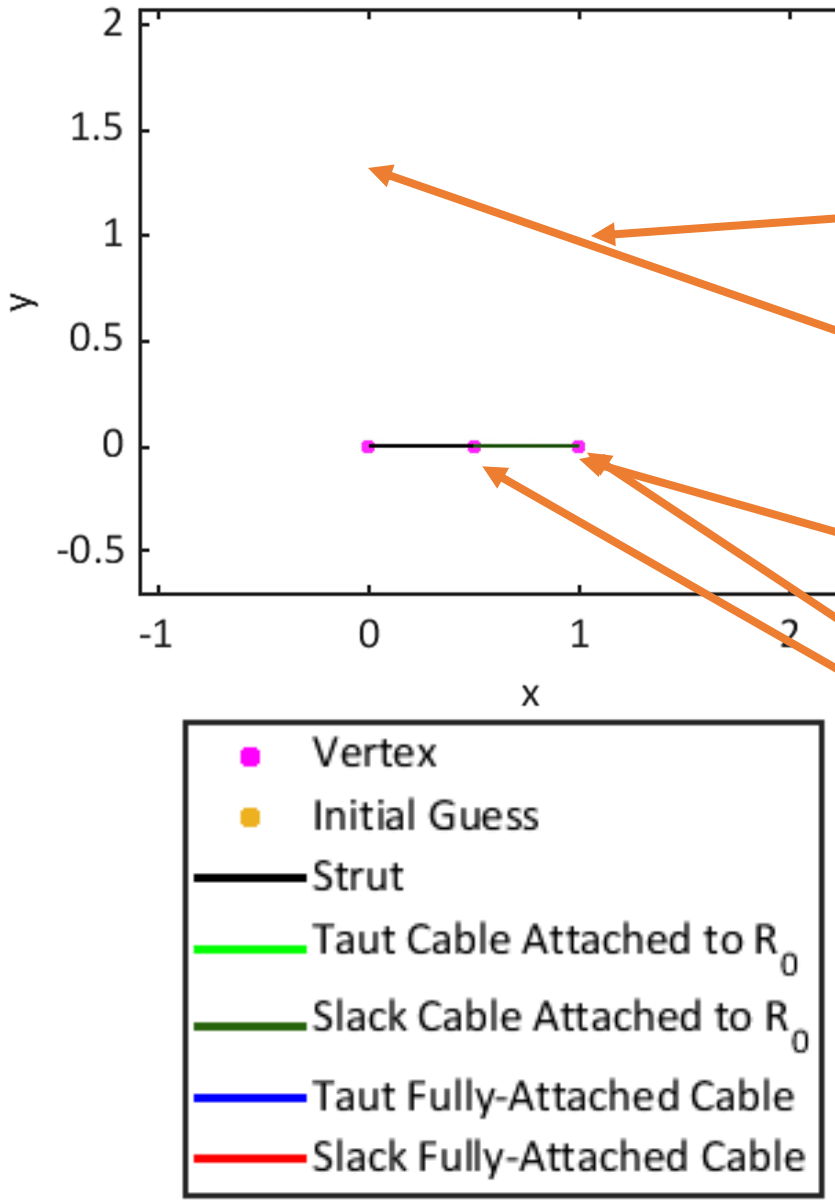


Slot to slide strut for testing of partial ejection from R_2

Caltech

Octagon (No Angle Stops)

$R_0 = 0.5L$ MATLAB, No Convexity Constraint



Regular polygon due to slack cables

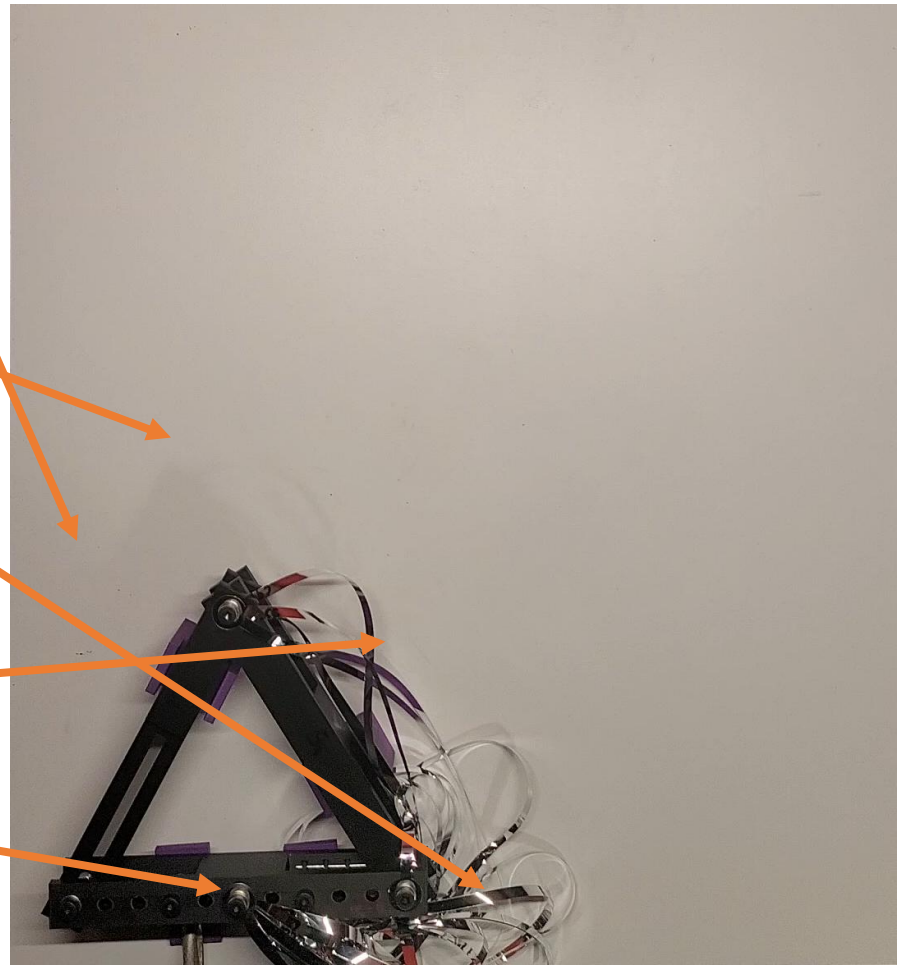
Irregular shape due to taut cables

Concave penultimate shape

Concave final shape

R_0

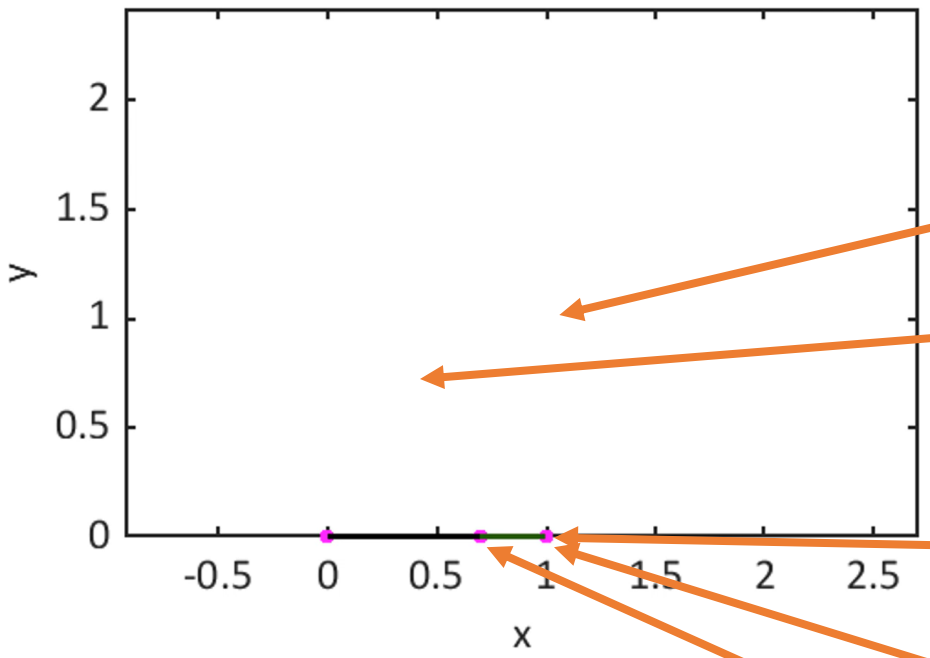
$R_0 = 0.5L$ Prototype, No Angle Stops

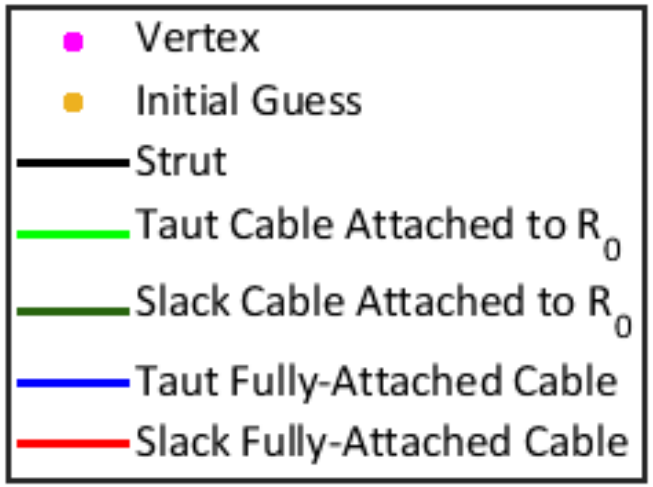


Caltech

Octagon (Angle Stops)

$R_0 = 0.7L$ MATLAB, Convexity Constraint





Regular polygon due to slack cables

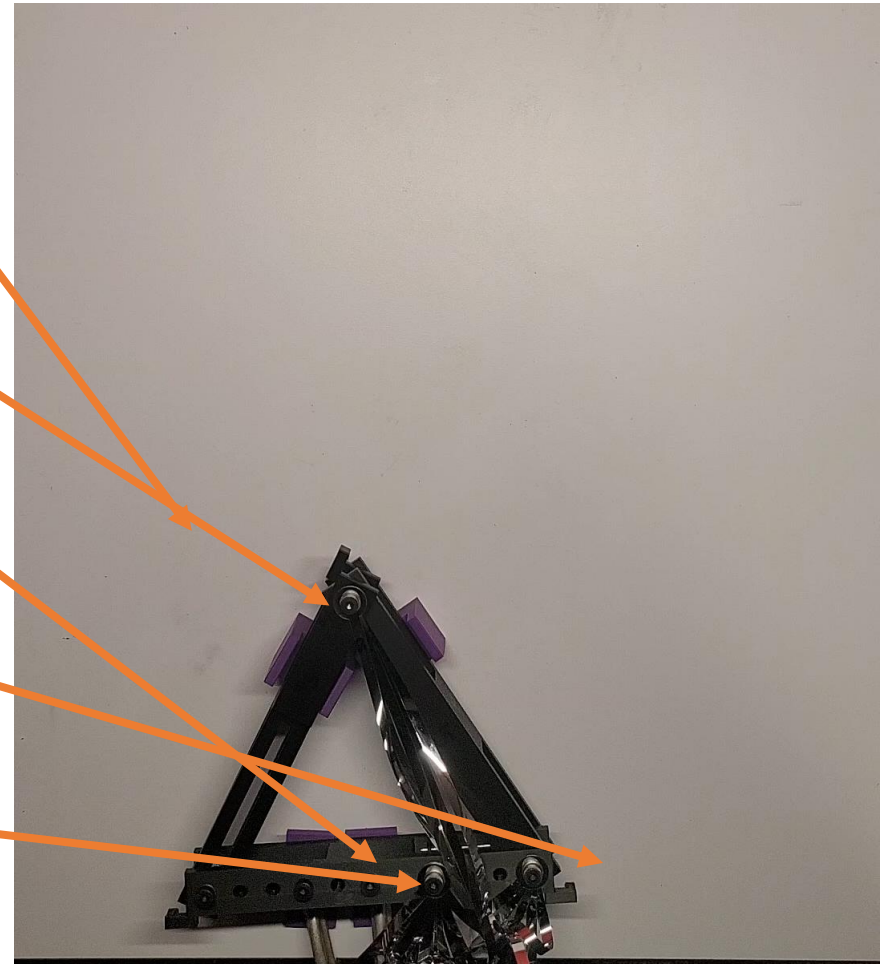
Irregular shape due to taut cables

Angle stops active

Desired regular octagon

R_0

$R_0 = 0.7L$ Prototype, Angle Stops



Caltech

Regular Polygon Feasibility Proof

Hypothesis

Setting $R_0 = R_2$ allows for a feasible sequence of regular polygons at all steps for any n, with the final step recovering the desired shape

Requirements for Feasibility at Each Step:

- All struts have length L (automatically satisfied by regular polygons)
- All angles satisfy convexity constraint (automatically satisfied by regular polygons)
- All cables have length less than or equal to taut length (need to prove)

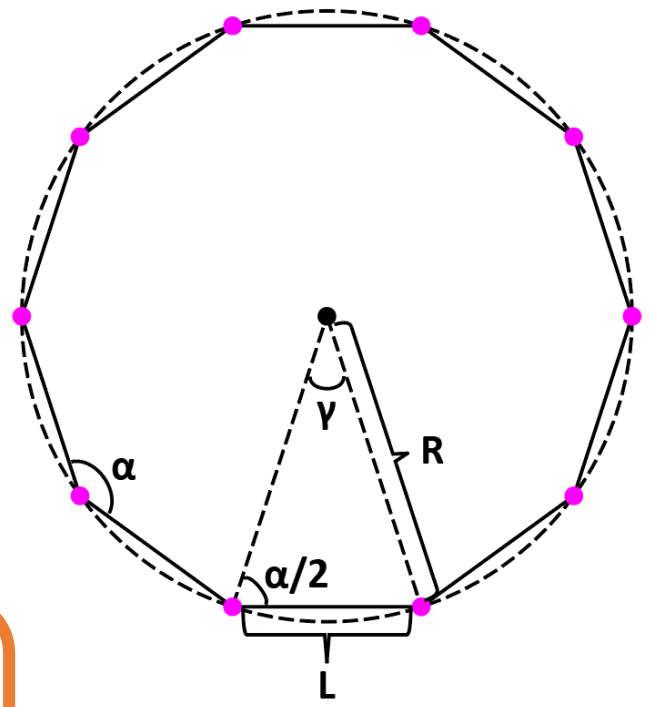
To Prove Cable Constraint Satisfaction:

- Calculate taut cable lengths from diagonal lengths of final regular polygon (so final shape automatically satisfies all constraints)
- Demonstrate that fully-attached cables at intermediate steps satisfy cable constraints
- Demonstrate that cables attached to R_0 at intermediate steps satisfy cable constraints

Regular Polygon Feasibility Proof

- Calculation of cable taut lengths:
  - Interior angles in final regular n-sided polygon: $\alpha = \pi(n-2)/n$
  - Central angle of circumscribed circle subtended by chord representing 1 side of polygon: $\gamma = 2\pi/n$
  - Law of sines for isosceles triangle connecting center of circle with 1 side of polygon: $\frac{\sin(\alpha/2)}{R} = \frac{\sin(\gamma)}{L}$
  - Solve for radius of circumscribed circle and simplify trigonometric identities: $R = \frac{L}{2\sin(\pi/n)}$
  - Central angle subtended by chord connecting across k-vertex separation: $\beta = 2\pi k/n$

Final Regular Polygon Schematic



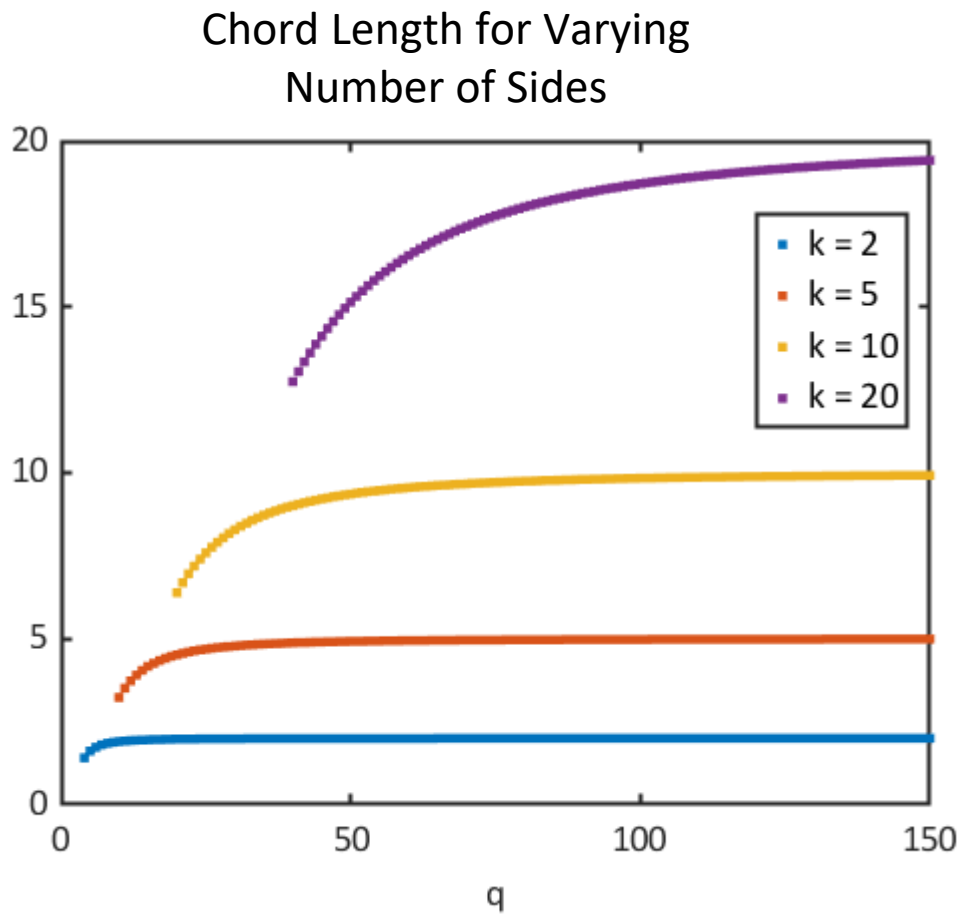
Taut length of cable with k-vertex separation in an n-sided polygon with side length L (based on known equation for chord length of a circle):

$$c(n,k,L) = 2R\sin(\beta/2) = \frac{L\sin(\pi k/n)}{\sin(\pi/n)}$$

Regular Polygon Feasibility Proof

- Fully-attached cable constraint satisfaction at intermediate steps:
  - Length of fully-attached cable across k-vertex separation in q-sided intermediate polygon: $c(q;k,L) = \frac{L\sin(\pi k/q)}{\sin(\pi/q)}$
  - A q-sided intermediate polygon has chords with vertex separations $k \in \{2,3,\dots,\text{floor}(q/2)\}$, so chords with a given k-vertex separation only exist for $q \geq 2k$
  - Plotting $c(q;k,L)$ vs. q for various k and fixed L demonstrates that c increases monotonically with q (horizontal asymptote at $c = Lk$)

A fully-attached cable in an intermediate q-sided polygon will have a length less than or equal to its taut length in the final n-sided polygon for $q \leq n$



Regular Polygon Feasibility Proof

- R_0-attached cable constraint satisfaction at intermediate steps:
  - Cable $C_{r,s}$ connects V_r to V_s, where $s - r > 1$
  - $C_{r,s}$ is attached to $R_0 = R_2 = V_q$ at a step forming a q-sided polygon if V_r has been ejected and V_s has not, i.e. $r < q \leq s$
  - Need to show that vertex separation between V_r and V_q is less than or equal to final k-vertex separation in same direction as separation between V_r and V_s

Counterclockwise:

- Counterclockwise vertex separation from V_r to V_s is $s - r = k$
- Counterclockwise vertex separation from V_r to V_q is $q - r$
- Subtract r from all terms of $r < q \leq s$: $0 < q - r \leq s - r$
- Vertex separation $\leq k$

Clockwise:

- Clockwise vertex separation from V_r to V_s is $(r + n) - s = k \rightarrow k - r = n - s$
- Clockwise vertex separation from V_r to V_q is r
- $r \leq k$ because $s \leq n$ (final vertex in final polygon is V_n)
- Vertex separation $\leq k$

Regular Polygon Feasibility Proof

- R_0-attached cable constraint satisfaction at intermediate steps:
  - Need to show that cable length increases monotonically with vertex separation, so if separation $< k$ due to R_0 attachment, cable has more slack than if it were fully connected
    - Cable length for $(k-1)$-vertex separation: $c(q, k-1, L) = c(q, k, L)\cos(\pi/q) - \cos(\pi k/q)$
    - For integer $q \geq 3$ and $k \in \{2,3, \ldots, \text{floor}(q/2)\}$, $\cos(\pi/q) \in (0,1)$ and $\cos(\pi k/q) \in [0,1)$
    - Thus, $c(q, k-1, L) < c(q, k, L)$ (same logic holds recursively for $k-2$, $k-3$, etc.)

A cable attached to R_0 in an intermediate step will have a length less than or equal to its taut length in the final polygon due to: (1) vertex separation being less than or equal to final k-vertex separation, and (2) smaller vertex separations corresponding to cables with more slack

Conclusion

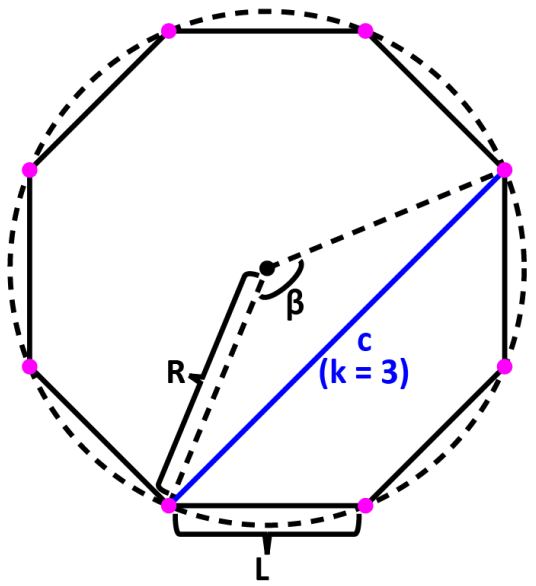
For an n-sided regular polygon with cables on all diagonals and $R_0 = R_2$, the set of q-sided regular polygons ($q \leq n$) satisfies all constraints at each step and therefore guarantees the existence of a successful assembly sequence for any n

Regular Polygon Feasibility: Fully-Attached Cables

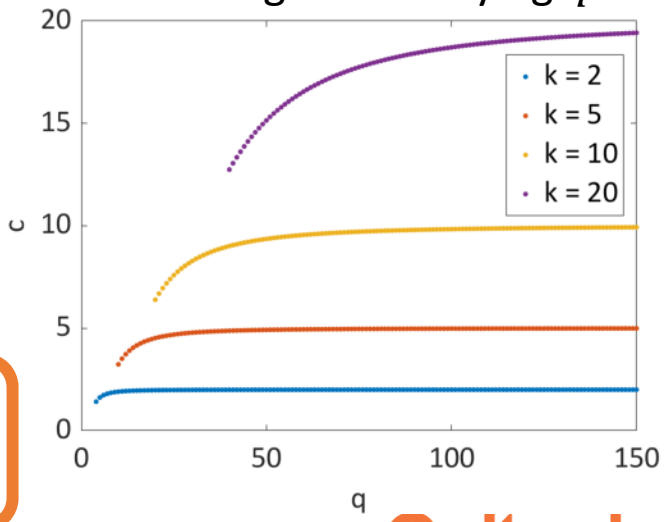
- Circumradius of n-sided regular polygon of side length L:
  - $R = \frac{L}{2\sin(\pi/n)}$
- Central angle subtended by chord across k vertices:
  - $\beta = 2\pi k/n$
- Taut length of cable with vertex separation k:
  - $c(n,k,L) = 2R\sin(\beta/2) = \frac{L\sin(\pi k/n)}{\sin(\pi/n)}$
- Length of fully-attached cable in q-sided intermediate polygon:
  - $c(q,k,L) = \frac{L\sin(\pi k/q)}{\sin(\pi/q)}$
- c increases monotonically with q for fixed L and $k \in [2, \text{floor}\left(\frac{q}{2}\right)]$
  - For $q < n$, $c(q,k,L) < c(n,k,L)$

Fully-attached cables satisfy length constraints at intermediate steps

Final Regular Polygon



Cable Length for Varying q



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Regular Polygon Feasibility: Cables Attached to R_0

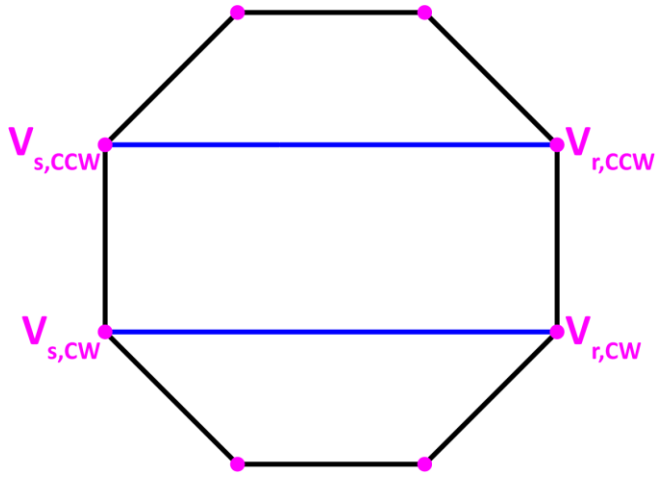
- Cable $\vec{C}_{r,s}$ is attached to $R_0 = V_q$ if $r < q \leq s$
- For a cable with a final k-vertex separation:
  - V_r to R_0 separation $\leq k$ at earlier steps
  - Holds for clockwise and counterclockwise separations
- Cable length for $(k-1)$-vertex separation:
  - $c(q, k-1, L) = c(q, k, L)\cos(\pi/q) - \cos(\pi k/q)$
- Due to bounds on cosine, $c(q, k-1, L) < c(q, k, L)$
  - Decreased vertex separation → Decreased cable length

Cables attached to R_0 satisfy length constraints at intermediate steps

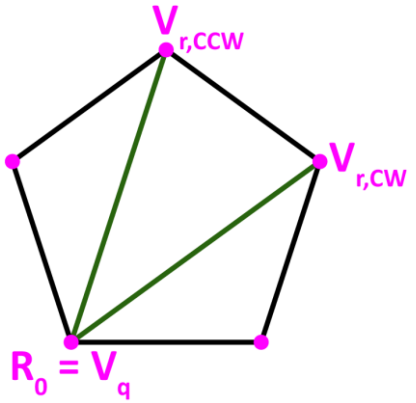
Conclusion

Sequence of regular polygons formed by $(R_0, R_1) = (0, L)$ always satisfies cable length constraints: Fully feasible for any n

Cables with $k = 3$ in Final Polygon



R_0 Attachment in Intermediate Shape



Caltech

Regular Polygon Feasibility Proof: Cables Attached to R_0

- Cable $\vec{C}_{r,s}$ is attached to $R_0 = V_q$ if $r < q \leq s$

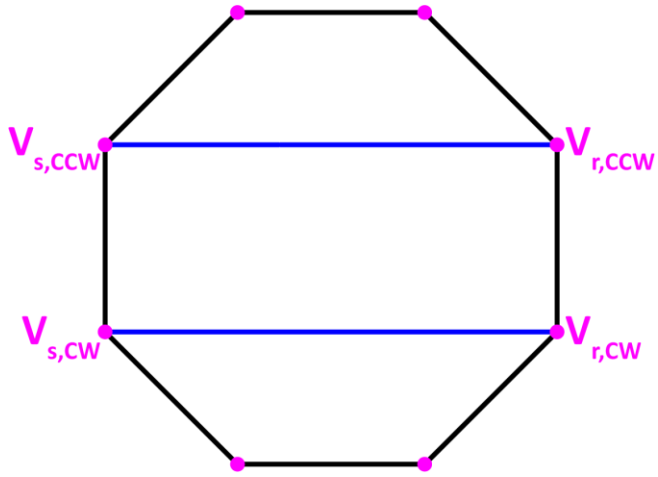
Counterclockwise (CCW):

- Separation k: $s - r = k$
- Separation from V_r to R_0: $q - r$
- $q \leq s$ implies $q - r \leq s - r$: Separation $\leq k$

Clockwise (CW):

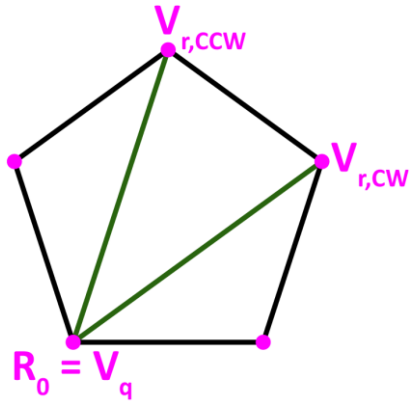
- Separation k: $(r + n) - s = k$ → $k - r = n - s$
- Separation from V_r to R_0: r
- $s \leq n$ implies $r \leq k$: Separation $\leq k$

Cables with $k = 3$ in Final Polygon



- Cable length for $(k-1)$-vertex separation:
  - $c(q, k-1, L) = c(q, k, L)\cos(\pi/q) - \cos(\pi k/q)$
  - Due to bounds on range of cosine, $c(q, k-1, L) < c(q, k, L)$

R_0 Attachment in Intermediate Shape



Vertex separation from V_r to $R_0 \leq$ Final separation

Decreased separation implies decreased cable length

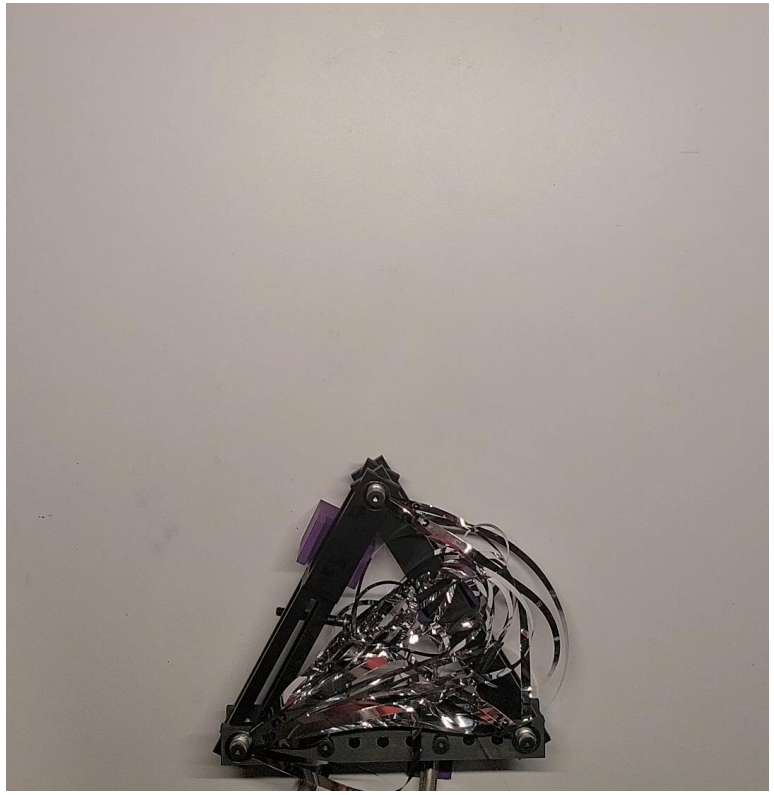
Cables attached to R_0 satisfy length constraints at intermediate steps

Caltech

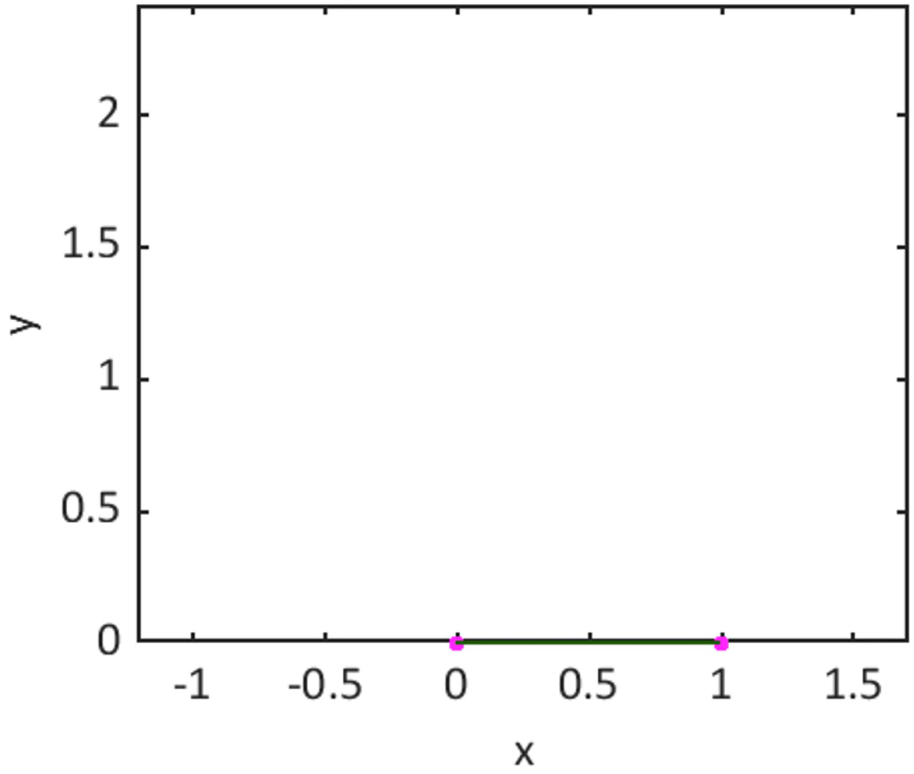
$R_0 = 0$ Octagon Videos

- Regular polygons at each step, as expected
- Prototype requires repeated attachment/detachment of cables from $R_0 = R_2$
  - Demonstrates practical difficulties associated with having parking spot coincide with strut/cable attachment and ejection location

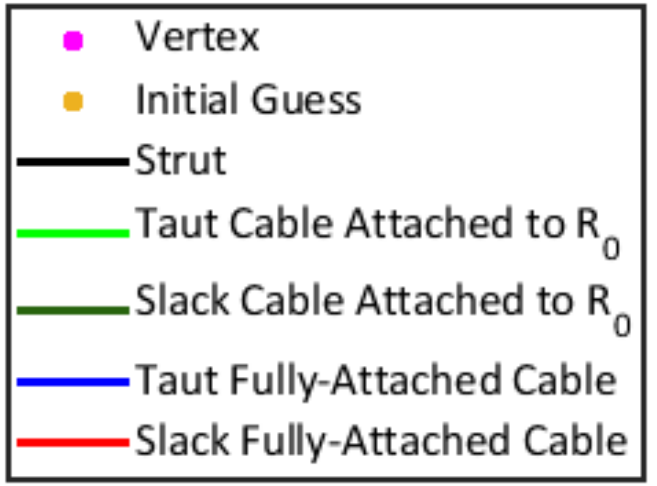
Prototype



MATLAB Model



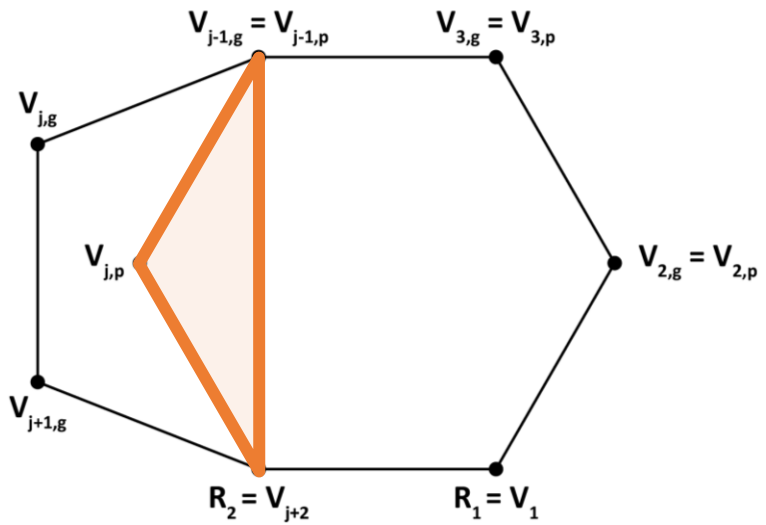
MATLAB Model Legend



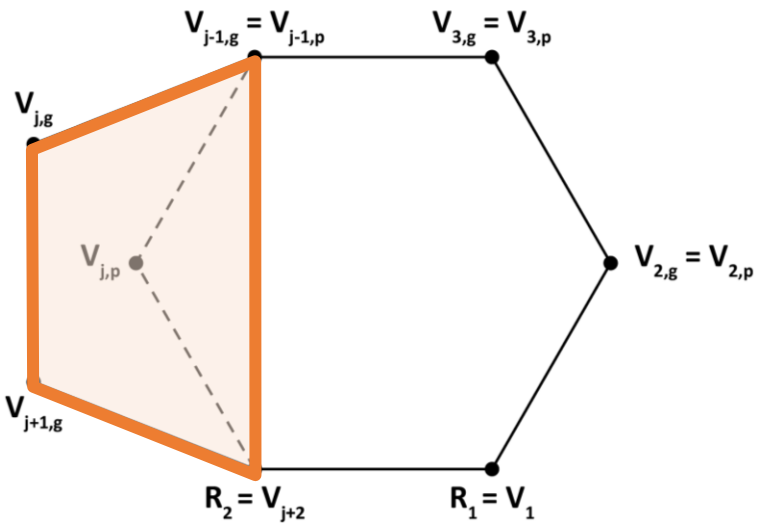
Optimization Guess Overview

- 1st step guess = hardcoded, last step guess = penultimate shape
- For Step j, take guesses for V_2 to V_{j-1} as positions in converged solution from Step $j-1$
- Convert triangle between $V_{j-1,p}$, R_2, and $V_{j,p}$ into a trapezoid between $V_{j-1,g}$, R_2, and guesses of V_j and V_{j+1}
- Trapezoid should have one base and both legs of length L to satisfy strut length constraints (brings guess close to feasible region of optimization problem)
- Cable length constraints may not be satisfied by guess – optimization is responsible for locating a feasible solution near guess if one exists

Shape from Step $j-1$



Guess for Step j

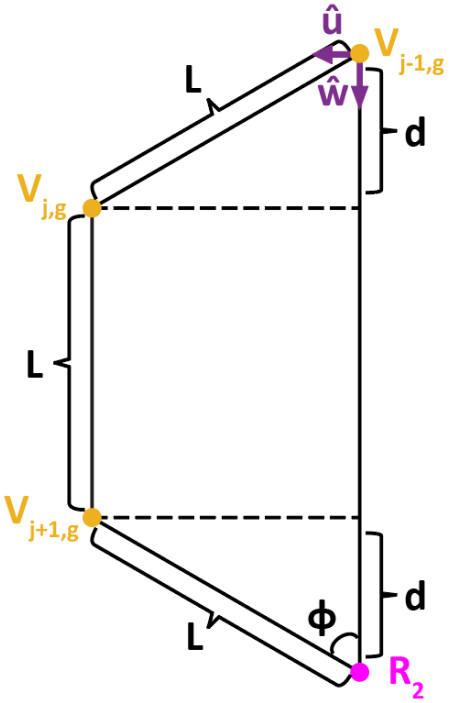


Caltech

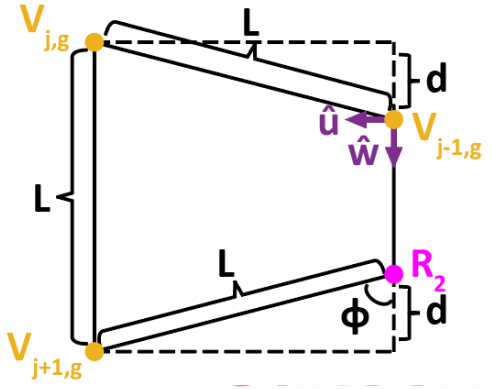
Optimization Guess Details

- Positions of R_2 and $V_{j-1,g}$ are known from prior step
- Need to find $V_{j,g}$ and $V_{j+1,g}$ based on distances parallel/perpendicular to line connecting R_2 to $V_{j-1,g}$
- Define geometric parameters:
  - Length $d = \left|0.5\left(\left|R_2 - V_{j-1,g}\right| - L\right)\right|$
  - Angle $\phi = \arccos(d/L)$
  - Orthogonal unit vectors $\hat{w} = \frac{R_2 - V_{j-1,g}}{\left|R_2 - V_{j-1,g}\right|}$ and $\hat{u} = \left((\hat{w})_y, -(\hat{w})_x\right)$
- If $\left|R_2 - V_{j-1,g}\right| > L$:
  - $V_{j,g} = V_{j-1,g} + d\hat{w} + L\sin(\phi)\hat{u}$
  - $V_{j+1,g} = V_{j-1,g} + (L + d)\hat{w} + L\sin(\phi)\hat{u}$
- If $\left|R_2 - V_{j-1,g}\right| \leq L$:
  - $V_{j,g} = V_{j-1,g} - d\hat{w} + L\sin(\phi)\hat{u}$
  - $V_{j+1,g} = V_{j-1,g} + (L - d)\hat{w} + L\sin(\phi)\hat{u}$

$\left|R_2 - V_{j-1}\right| > L$



$\left|R_2 - V_{j-1}\right| \leq L$



(R_0, R_1) Grid Searches

- Clear "staircase" structure of feasible cases
- As n increases, number of valid cases (and corresponding maximum R_0) decreases
  - Any $R_0 > 0.5L$: $n = 8$
  - $R_0 = 0.5L$: $n = 9$
  - $R_0 = 0.4L$: $n = 10$
  - $R_0 = 0.3L$: $n = 12$
  - $R_0 = 0.2L$: $n = 15$
  - $R_0 = 0.1L$: $n = 23$
- For all $n > 23$, only $R_0 = 0$ is feasible
- Grid searches are discrete – if mesh is refined further, may be able to assemble larger n for some $R_0 > 0$

